TOPIC 4 • THEORY Advanced Mathematics • BSc Aerospace Engineering • UC3M

Systems of First-Order ODEs and the Phase Plane

Pages 9 and 10 of the class notes, in their order: systems and their existence theorem, linear systems with the Wronskian and the fundamental set, elimination and the eigenvalue method, and the three cases of a 2×2 matrix drawn by a change of basis $\mathbf{z} = P^{-1}\mathbf{x}$ – the six orderings of two real eigenvalues and a zero eigenvalue, complex eigenvalues through the complex diagonalisation of the notes, and a repeated eigenvalue with the construction of the generalised eigenvector. Robinson material beyond the notes is labelled as a complement.

Ground truth: Class notes Notes.pdf, pp. 9-10 • Robinson (2004) Ch. 25 (pp. 249-252), Ch. 26 (pp. 253-258), Ch. 27 (pp. 259-268), Ch. 28 (pp. 269-284), Ch. 29 (pp. 285-293), Ch. 30 (pp. 295-300), Ch. 31 (pp. 301-305), Appendix B (pp. 382-385) • Lecture map, Part IV (Oct 6 - Oct 15)

≈ 70 min study time • 71 displayed equations • 16 figures

SECTION 01 • NOTES P. 9 • ROBINSON CH. 25 • LECTURE OCT 6

1. Vector equations and their theory

SYSTEMS OF DIFFERENTIAL EQUATIONS [NOTES, P. 9]

We have n dependent variables $x_1, x_2, \ldots, x_n$ and one independent variable t. They satisfy a system of ODEs, $\dot{x}_1 = f_1(x_1, \ldots, x_n, t),\ \ldots,\ \dot{x}_n = f_n(x_1, \ldots, x_n, t)$. Notation: $\mathbf{x} = (x_1, \ldots, x_n)^T$ is a function of t and $\mathbf{f} = (f_1, \ldots, f_n)^T$ a function of $\mathbf{x}$ and t, so that $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, t)$.

If $\mathbf{f}$ and $D\mathbf{f}$ (the Jacobian matrix) are continuous functions of $\mathbf{x}$ and t in an open neighbourhood of $\mathbf{x}_0$ and t_0, then the IVP $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, t), \mathbf{x}(t_0) = \mathbf{x}_0$ has a unique local solution.

Robinson states the same theory as follows.

A system of n coupled first-order equations [Robinson, Eq. (25.1), p. 249]

$$\dot{x}_i = f_i(x_1, \ldots, x_n, t), \qquad i = 1, \ldots, n, \tag{25.1}$$

is written compactly with $\mathbf{x} = (x_1, \ldots, x_n)^T$ and $\mathbf{f} = (f_1, \ldots, f_n)^T$ as

$$\frac{d\mathbf{x}}{dt} = \mathbf{f}(\mathbf{x}, t).$$

For $n = 1$ this is the scalar equation of Topic 2, and the theory is the same with vectors in place of numbers.

DEFINITION 25.1 – SOLUTION OF A VECTOR IVP [ROBINSON, P. 250]
A solution of $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, t), \mathbf{x}(t_0) = \mathbf{x}_0 \in \mathbb{R}^n$ (25.2) on an open interval $I \ni t_0$ is a differentiable function $\mathbf{x} : I \to \mathbb{R}^n$ with $\mathbf{x}(t_0) = \mathbf{x}_0$ and $\dot{\mathbf{x}}(t) = \mathbf{f}(\mathbf{x}(t), t)$ for all $t \in I$.

The role of $\partial f/\partial x$ is now played by the matrix of all partial derivatives, $D\mathbf{f}$ (usually called the **Jacobian**; Robinson just says "the matrix of partial derivatives") [Eq. (25.3), p. 250]:

$$D\mathbf{f} = \begin{pmatrix} \partial f_1/\partial x_1 & \cdots & \partial f_1/\partial x_n \\ \vdots & & \vdots \\ \partial f_n/\partial x_1 & \cdots & \partial f_n/\partial x_n \end{pmatrix}. \tag{25.3}$$

THEOREM 25.2 – EXISTENCE AND UNIQUENESS FOR SYSTEMS [ROBINSON, P. 250]
If $\mathbf{f}(\mathbf{x}, t)$ and $D\mathbf{f}(\mathbf{x}, t)$ are continuous for $\mathbf{x} \in U = (a_1, b_1) \times \cdots \times (a_n, b_n)$ and $c < t < d$, then for any $\mathbf{x}_0 \in U$ and $t_0 \in (c, d)$ the IVP (25.2) has a unique solution on some open interval containing t_0.

In practice: check that every f_i and every partial derivative $\partial f_i/\partial x_j$ is continuous.

EXAMPLE – THE COMPETING SPECIES OF TOPIC 1
For $\mathbf{f} = \big(x(4 - 2x - y),\ y(9 - 3x - 3y)\big)$, $D\mathbf{f} = \begin{pmatrix} 4 - 4x - y & -x \\ -3y & 9 - 3x - 6y \end{pmatrix}$. All entries are polynomials, hence continuous on all of $\mathbb{R}^2$: every initial condition has a unique local solution, and in particular two trajectories of Figure 1.1 of Topic 1 can never cross.

1.1 Higher-order equations as systems [Robinson, Sec. 25.1, pp. 251–252]

IMPORTANT POINT [NOTES, P. 9]
ODEs of order $n \geq 2$ can be converted into a system of n first-order ODEs. Example: $\ddot{x} = f(\dot{x}, x, t)$ becomes $\dot{x}_1 = x_2, \dot{x}_2 = f(x_2, x_1, t)$ with $x_1 = x, x_2 = \dot{x}$.

For $\ddot{x} = f(\dot{x}, x, t)$ introduce $x_1 = x$ and $x_2 = \dot{x}$. Then

$$\dot{x}_1 = x_2, \qquad \dot{x}_2 = \ddot{x} = f(x_2, x_1, t), \qquad \text{i.e.} \qquad \frac{d\mathbf{x}}{dt} = \mathbf{g}(\mathbf{x}, t) = \begin{pmatrix} x_2 \\ f(x_2, x_1, t) \end{pmatrix}. \tag{25.4}$$

The vector initial condition $\mathbf{x}(t_0) = \mathbf{x}_0$ is exactly the pair $x(t_0), \dot{x}(t_0)$, and $D\mathbf{g} = \begin{pmatrix} 0 & 1 \\ \partial f/\partial x_1 & \partial f/\partial x_2 \end{pmatrix}$. Applying Theorem 25.2 gives **Theorem 25.3**, which is Theorem 11.1 of Topic 3. An nth-order equation becomes an n-dimensional system in the same way, with $x_k = x^{(k-1)}$.

EXAMPLE — THE MASS-SPRING-DAMPER AS A SYSTEM

$\ddot{x} + \lambda\dot{x} + \omega^2 x = 0$ with $x_1 = x$ (position) and $x_2 = \dot{x}$ (velocity) becomes

$$\frac{d}{dt}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -\omega^2 & -\lambda \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}.$$

The plane of $(x, \dot{x})$ is the **phase plane**; we return to this example in Section 9.

2. Linear systems and the elimination method

A first-order linear system of two equations with constant coefficients is [Robinson, Eq. (26.1), p. 253]

$$\dot{x} = ax + by + f(t), \qquad \dot{y} = cx + dy + g(t). \tag{26.1}$$

It is **homogeneous** if $f = g = 0$ and **inhomogeneous** otherwise. If $b = c = 0$ the equations are uncoupled and each is solved by an integrating factor.

METHOD 1: ELIMINATION, THE EXAMPLE OF THE NOTES [NOTES, P. 9]

For $\dot{x} = ax + by$, $\dot{y} = cx + dy$ (with $b \neq 0$):

1. From the first equation, $y = \dfrac{\dot{x} - ax}{b}$, so $\dot{y} = \dfrac{\ddot{x} - a\dot{x}}{b}$.
2. Substitute into the second equation: $\dfrac{\ddot{x} - a\dot{x}}{b} = cx + d\,\dfrac{\dot{x} - ax}{b}$.
3. Multiply by b: $\ddot{x} - a\dot{x} = bc\,x + d\dot{x} - ad\,x$, that is, a **second-order** equation for x:

$$\ddot{x} - (a + d)\,\dot{x} + (ad - bc)\,x = 0.$$

The system $x' = 2x - y$, $y' = -x + 3y$ of Topic 1 gives $\ddot{x} - 5\dot{x} + 5x = 0$ this way.

Robinson's version also carries forcing terms:

2.1 Reduction to one second-order equation

Assume $b \neq 0$ (otherwise exchange the roles of x and y).

1. Solve the first equation for y: $y = \dfrac{\dot{x} - ax - f(t)}{b}$ (26.2).
2. Differentiate: $\dot{y} = \dfrac{\ddot{x} - a\dot{x} - \dot{f}(t)}{b}$.
3. Substitute both into the second equation: $\dfrac{\ddot{x} - a\dot{x} - \dot{f}}{b} = cx + d\,\dfrac{\dot{x} - ax - f}{b} + g$.
4. Multiply by b and collect terms:

$$\ddot{x} - (a + d)\,\dot{x} + (ad - bc)\,x = \dot{f}(t) + b\,g(t) - d\,f(t). \tag{26.3}$$

Solve (26.3) by the methods of Topic 3, then recover y from (26.2). The coefficients $a + d$ and $ad - bc$ are the trace and the determinant of the matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ – they will reappear in the eigenvalue method.

EXAMPLE 26.1 (PP. 254-255): $\dot{x} = x + y,\ \dot{y} = 4x - 2y + 4e^{-2t},\ x(0) = 0,\ y(0) = -1$

1. Here $a = 1, b = 1, c = 4, d = -2, f = 0, g = 4e^{-2t}$. From the first equation $y = \dot{x} - x$ (26.4), so $\dot{y} = \ddot{x} - \dot{x}$.
2. Substituting: $\ddot{x} - \dot{x} = 4x - 2(\dot{x} - x) + 4e^{-2t}$, i.e. $\ddot{x} + \dot{x} - 6x = 4e^{-2t}$ (26.5) – as (26.3) predicts: $a + d = -1$, $ad - bc = -6$, right side $b\,g = 4e^{-2t}$.
3. Complementary function: $k^2 + k - 6 = 0, k = 2, -3$. Particular integral Ce^{-2t}: $(4 - 2 - 6)C = 4, C = -1$. So $x = Ae^{2t} + Be^{-3t} - e^{-2t}$.
4. $y = \dot{x} - x = (2A - A)e^{2t} + (-3B - B)e^{-3t} + (2 + 1)e^{-2t} = Ae^{2t} - 4Be^{-3t} + 3e^{-2t}$.
5. Initial data: $A + B - 1 = 0$ and $A - 4B + 3 = -1$. Subtracting, $5B = 5$: $B = 1, A = 0$.

$$x(t) = e^{-3t} - e^{-2t}, \qquad y(t) = 3e^{-2t} - 4e^{-3t}.$$

Check: $\dot{x} = -3e^{-3t} + 2e^{-2t}$ and $x + y = 2e^{-2t} - 3e^{-3t}$; $x(0) = 0, y(0) = -1$.

EXAMPLE 26.2 (PP. 255-256): $\dot{x} = 2x + 5y,\ \dot{y} = -2x,\ x(0) = 5,\ y(0) = -4$

$y = \frac{1}{5}(\dot{x} - 2x)$, so $\frac{1}{5}(\ddot{x} - 2\dot{x}) = -2x$, i.e. $\ddot{x} - 2\dot{x} + 10x = 0$, with roots $k = 1 \pm 3i$: $x = e^{t}(A\cos 3t + B\sin 3t)$ and $y = \frac{e^{t}}{5}\left[(3B - A)\cos 3t - (3A + B)\sin 3t\right]$. The data give $A = 5, B = -5$: $x = 5e^{t}(\cos 3t - \sin 3t)$, $y = -e^{t}(4\cos 3t + 2\sin 3t)$ – oscillations with exponentially growing amplitude.

3. Matrix form, superposition and the fundamental matrix

The homogeneous system $\dot{x} = ax + by$, $\dot{y} = cx + dy$ (27.1) is, with $\mathbf{x} = (x, y)^T$ [Eq. (27.2), p. 259],

$$\dot{\mathbf{x}} = A\mathbf{x}, \qquad A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}. \tag{27.2}$$

SUPERPOSITION [ROBINSON, EQ. (27.3), P. 260]

If $\mathbf{x}_1$ and $\mathbf{x}_2$ solve (27.2), so does $\mathbf{x} = \alpha\mathbf{x}_1 + \beta\mathbf{x}_2$: $\dot{\mathbf{x}} = \alpha\dot{\mathbf{x}}_1 + \beta\dot{\mathbf{x}}_2 = \alpha A\mathbf{x}_1 + \beta A\mathbf{x}_2 = A(\alpha\mathbf{x}_1 + \beta\mathbf{x}_2) = A\mathbf{x}$.

SPECIAL IMPORTANT CASE: LINEAR SYSTEMS [NOTES, P. 9]

$\dot{\mathbf{x}} = A(t)\,\mathbf{x} + \mathbf{f}(t), \mathbf{x}(t_0) = \mathbf{x}_0$, where A is an $n \times n$ matrix (depending on t) and $\mathbf{f}$ a vector function of t; if $\mathbf{f} = \mathbf{0}$ the system is homogeneous. Existence and uniqueness are guaranteed if A and $\mathbf{f}$ are continuous.

- **Notation.** For two solutions write $\mathbf{x}_1 = (x_{11}, x_{21}, \ldots, x_{n1})^T, \mathbf{x}_2 = (x_{12}, x_{22}, \ldots, x_{n2})^T$.
- **Superposition.** If $\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_k$ solve $\dot{\mathbf{x}} = A(t)\mathbf{x}$, so does every linear combination $c_1\mathbf{x}_1 + c_2\mathbf{x}_2 + \cdots + c_k\mathbf{x}_k$.
- **Theorem.** The solutions $\mathbf{x}_1, \ldots, \mathbf{x}_k$ of $\dot{\mathbf{x}} = A(t)\mathbf{x}$ are linearly independent if and only if the Wronskian $W(\mathbf{x}_1, \ldots, \mathbf{x}_k) = \det\left(x_{ij}\right) \neq 0$ for all $t \in I$.
- $n = 2$. $\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \end{pmatrix} = \begin{pmatrix} a_{11}(t) & a_{12}(t) \\ a_{21}(t) & a_{22}(t) \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} f_1(t) \\ f_2(t) \end{pmatrix}$. A **fundamental set** of solutions is a pair $\mathbf{x}_1(t)$, $\mathbf{x}_2(t)$ with $W(\mathbf{x}_1, \mathbf{x}_2) \neq 0$.
- **Theorem.** The general solution of $\dot{\mathbf{x}} = A(t)\mathbf{x} + \mathbf{f}(t)$ is $\mathbf{x} = \mathbf{x}_h + \mathbf{x}_p$, where $\mathbf{x}_h$ is the general solution of $\dot{\mathbf{x}} = A(t)\mathbf{x}$ and $\mathbf{x}_p$ a particular solution of the complete system.

Robinson does not develop this general theory for systems; the nearest book material is the scalar version (Sec. 11.3–11.4 and 14.1). The statements below make the notes precise for constant A; the proofs mirror Topic 3.

DEFINITIONS — INDEPENDENCE, WRONSKIAN, FUNDAMENTAL MATRIX

- Vector functions $\mathbf{x}_1(t), \mathbf{x}_2(t)$ are **linearly independent** on I if $\alpha\mathbf{x}_1(t) + \beta\mathbf{x}_2(t) = \mathbf{0}$ for all $t \in I$ forces $\alpha = \beta = 0$.
- Their **Wronskian** is the determinant of the matrix whose columns are the solutions: $W(t) = \det\left[\mathbf{x}_1(t) \;\; \mathbf{x}_2(t)\right] = x_{11}x_{22} - x_{12}x_{21}$, where $\mathbf{x}_j = (x_{1j}, x_{2j})^T$.
- Two independent solutions form a **fundamental set**; the matrix $\Phi(t) = [\mathbf{x}_1 \;\; \mathbf{x}_2]$ is a **fundamental matrix**. It satisfies $\dot{\Phi} = A\Phi$, column by column.

THEOREM — WRONSKIAN TEST AND GENERAL SOLUTION FOR SYSTEMS

1. Two solutions of $\dot{\mathbf{x}} = A\mathbf{x}$ are linearly independent if and only if $W(t) \neq 0$; and then $W(t) \neq 0$ for all t.
2. If Φ is a fundamental matrix, every solution is $\mathbf{x}(t) = \Phi(t)\,\mathbf{c} = c_1\mathbf{x}_1(t) + c_2\mathbf{x}_2(t)$, and the IVP $\mathbf{x}(t_0) = \mathbf{x}_0$ has $\mathbf{c} = \Phi(t_0)^{-1}\mathbf{x}_0$.

PROOF

If $W(t_0) \neq 0$, the columns $\mathbf{x}_1(t_0), \mathbf{x}_2(t_0)$ are independent vectors, so $\alpha\mathbf{x}_1(t_0) + \beta\mathbf{x}_2(t_0) = \mathbf{0}$ already forces $\alpha = \beta = 0$. Conversely, if $W(t_0) = 0$, there are $(\alpha, \beta) \neq (0,0)$ with $\alpha\mathbf{x}_1(t_0) + \beta\mathbf{x}_2(t_0) = \mathbf{0}$. Then $\mathbf{z} = \alpha\mathbf{x}_1 + \beta\mathbf{x}_2$ solves the system (superposition) with $\mathbf{z}(t_0) = \mathbf{0}$; so does $\mathbf{0}$, and by uniqueness (Theorem 25.2) $\mathbf{z} \equiv \mathbf{0}$: the solutions are dependent. For (2): given any solution $\mathbf{x}$, the system $\Phi(t_0)\mathbf{c} = \mathbf{x}(t_0)$ is solvable because $\det \Phi(t_0) = W(t_0) \neq 0$; then $\Phi\mathbf{c}$ and $\mathbf{x}$ are solutions with the same value at t_0, hence equal for all t.

■

Liouville's formula – Abel's identity for systems (complementary)

Write the columns as $\mathbf{x}_j = (p_j, q_j)^T$, so $\dot p_j = ap_j + bq_j$, $\dot q_j = cp_j + dq_j$ and $W = p_1q_2 - q_1p_2$. By the product rule

$$\dot W = \dot p_1 q_2 + p_1 \dot q_2 - \dot q_1 p_2 - q_1 \dot p_2 = (ap_1 + bq_1)q_2 + p_1(cp_2 + dq_2) - (cp_1 + dq_1)p_2 - q_1(ap_2 + bq_2).$$

The b terms ($bq_1q_2 - bq_1q_2$) and the c terms ($cp_1p_2 - cp_1p_2$) cancel, leaving

$$\dot W = (a + d)(p_1q_2 - q_1p_2) = (\operatorname{tr} A)\, W \quad \Longrightarrow \quad W(t) = W(t_0)\, e^{(\operatorname{tr} A)(t - t_0)}.$$

(The computation never uses that a, b, c, d are constant; with time-dependent entries $W(t) = W(t_0) \exp \int_{t_0}^{t} \operatorname{tr} A(s)\, ds$.) For a second-order equation turned into a system, $A = \begin{pmatrix} 0 & 1 \\ -q & -p \end{pmatrix}$ has trace $-p$, the solution columns are $(x_j, \dot x_j)^T$, and W is exactly the scalar Wronskian $x_1\dot x_2 - x_2\dot x_1$: Abel's identity of Topic 3 is the special case.

3.1 Inhomogeneous systems

LEMMA — GENERAL SOLUTION OF $\dot{\mathbf{x}} = A\mathbf{x} + \mathbf{f}(t)$ (LECTURE; SCALAR ANALOGUE ROBINSON SEC. 14.1)

If $\mathbf{x}_p$ is one solution, every solution is $\mathbf{x} = \Phi(t)\mathbf{c} + \mathbf{x}_p(t)$.

PROOF

If $\mathbf{x}$ solves the inhomogeneous system, then $\frac{d}{dt}(\mathbf{x} - \mathbf{x}_p) = (A\mathbf{x} + \mathbf{f}) - (A\mathbf{x}_p + \mathbf{f}) = A(\mathbf{x} - \mathbf{x}_p)$, so $\mathbf{x} - \mathbf{x}_p$ solves the homogeneous system and equals $\Phi\mathbf{c}$. Conversely $\frac{d}{dt}(\Phi\mathbf{c} + \mathbf{x}_p) = A\Phi\mathbf{c} + A\mathbf{x}_p + \mathbf{f}$.

■

Variation of parameters for systems (complementary). As in Topic 3, let the constants vary: $\mathbf{x}_p = \Phi(t)\mathbf{c}(t)$. The product rule gives $\dot{\mathbf{x}}_p = \dot{\Phi}\mathbf{c} + \Phi\dot{\mathbf{c}} = A\Phi\mathbf{c} + \Phi\dot{\mathbf{c}}$; this equals $A\mathbf{x}_p + \mathbf{f}$ exactly when $\Phi\dot{\mathbf{c}} = \mathbf{f}$. Since $\det\Phi = W \neq 0$:

$$\mathbf{x}_p(t) = \Phi(t)\int \Phi(s)^{-1}\,\mathbf{f}(s)\,ds.$$

EXAMPLE 26.1 AGAIN, BY A FUNDAMENTAL MATRIX

The homogeneous solutions are $e^{2t}(1,1)^T$ and $e^{-3t}(1,-4)^T$ (Section 5), so $\Phi = \begin{pmatrix} e^{2t} & e^{-3t} \\ e^{2t} & -4e^{-3t} \end{pmatrix}$, $W = -4e^{-t} - e^{-t} = -5e^{-t}$ (check: $\operatorname{tr}A = -1$, so $W \propto e^{-t}$), and $\Phi^{-1} = \frac{1}{W}\begin{pmatrix} -4e^{-3t} & -e^{-3t} \\ -e^{2t} & e^{2t} \end{pmatrix} = \frac{1}{5}\begin{pmatrix} 4e^{-2t} & e^{-2t} \\ e^{3t} & -e^{3t} \end{pmatrix}$. With $\mathbf{f} = (0, 4e^{-2t})^T$: $\Phi^{-1}\mathbf{f} = \frac{4}{5}\left(e^{-4t},\ -e^{t}\right)^T$, whose antiderivative is $\left(-\frac{1}{5}e^{-4t},\ -\frac{4}{5}e^{t}\right)^T$. Then

$$\mathbf{x}_p = \Phi\begin{pmatrix} -\frac{1}{5}e^{-4t} \\ -\frac{4}{5}e^{t} \end{pmatrix} = \begin{pmatrix} -\frac{1}{5}e^{-2t} - \frac{4}{5}e^{-2t} \\ -\frac{1}{5}e^{-2t} + \frac{16}{5}e^{-2t} \end{pmatrix} = \begin{pmatrix} -e^{-2t} \\ 3e^{-2t} \end{pmatrix},$$

the same particular solution that elimination produced .

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4. The eigenvalue method

For the scalar equation $\dot{x} = ax$ the solutions are $e^{at}x_0$. By analogy try a solution that keeps a fixed direction $\mathbf{v}$ and changes only in size [Eq. (27.4), p. 260]:

$$\mathbf{x}(t) = e^{\lambda t}\mathbf{v}, \qquad \mathbf{v}\ \text{constant},\ \mathbf{v} \neq \mathbf{0}. \tag{27.4}$$

Then $\dot{\mathbf{x}} = \lambda e^{\lambda t}\mathbf{v}$ and $A\mathbf{x} = e^{\lambda t}A\mathbf{v}$; equating and dividing by $e^{\lambda t} \neq 0$:

$$A\mathbf{v} = \lambda\mathbf{v}. \tag{27.5}$$

So $e^{\lambda t}\mathbf{v}$ is a solution exactly when λ is an **eigenvalue** of A and $\mathbf{v}$ a corresponding **eigenvector**.

METHOD 2: LINEAR ALGEBRA [NOTES, P. 9]

Look for solutions of $\dot{\mathbf{x}} = A\mathbf{x}$ of the form $\mathbf{x}(t) = e^{\lambda t}\mathbf{v}$, with $\mathbf{v} \neq \mathbf{0}$ a vector and λ a number, both to be determined. Then $\lambda e^{\lambda t}\mathbf{v} = Ae^{\lambda t}\mathbf{v}$, and cancelling $e^{\lambda t}$, $A\mathbf{v} = \lambda\mathbf{v}$: the same as an eigenvalue problem, with $\mathbf{v}$ the associated eigenvector. We need to solve $\det(A - \lambda I) = 0$.

4.1 Finding eigenvalues and eigenvectors [Sec. 27.2, pp. 260–261]

1. Rewrite (27.5) as $(A - \lambda I)\mathbf{v} = \mathbf{0}$ (27.7). If $A - \lambda I$ were invertible, the only solution would be $\mathbf{v} = \mathbf{0}$. A non-zero $\mathbf{v}$ therefore requires $\det(A - \lambda I) = 0$.

2. Expand: $\det \begin{pmatrix} a-\lambda & b \\ c & d-\lambda \end{pmatrix} = (a-\lambda)(d-\lambda) - bc$, giving the **characteristic equation**

$$\lambda^2 - (a+d)\lambda + (ad - bc) = 0, \qquad \text{i.e.} \qquad \lambda^2 - (\operatorname{tr} A)\lambda + \det A = 0. \quad (27.8)$$

3. For each root λ, solve the (singular) system $(A - \lambda I)\mathbf{v} = \mathbf{0}$: one row is a multiple of the other, so one equation fixes the direction of $\mathbf{v}$. Any non-zero multiple is also an eigenvector, $A(c\mathbf{v}) = c\lambda\mathbf{v}$.

Equation (27.8) is the characteristic equation (26.3) of the eliminated second-order equation: both methods see the same numbers. Its roots satisfy $\lambda_1 + \lambda_2 = \operatorname{tr} A$ and $\lambda_1\lambda_2 = \det A$.

EXAMPLE 27.1 (PP. 262–263), THE EXAMPLE OF THE NOTES: $A = \begin{pmatrix} 2 & 2 \\ 6 & 3 \end{pmatrix}$ [NOTES, P. 9]

1. $(2-\lambda)(3-\lambda) - 12 = \lambda^2 - 5\lambda - 6 = (\lambda - 6)(\lambda + 1) = 0$: $\lambda_1 = 6, \lambda_2 = -1$. (Check: sum $5 = \operatorname{tr} A$, product $-6 = \det A$.)

2. $\lambda = 6$: $(A - 6I)\mathbf{v} = \begin{pmatrix} -4 & 2 \\ 6 & -3 \end{pmatrix}\mathbf{v} = \mathbf{0}$ gives $-4v_1 + 2v_2 = 0$, $v_2 = 2v_1$: $\mathbf{v}_1 = (1, 2)^T$. Check: $A\mathbf{v}_1 = (6, 12)^T = 6\mathbf{v}_1$.

3. $\lambda = -1$: $\begin{pmatrix} 3 & 2 \\ 6 & 4 \end{pmatrix}\mathbf{v} = \mathbf{0}$ gives $3v_1 + 2v_2 = 0$: $\mathbf{v}_2 = (2, -3)^T$. Check: $A\mathbf{v}_2 = (-2, 3)^T = -\mathbf{v}_2$.

TRIANGULAR MATRICES (EXAMPLES 27.2–27.3, P. 264)

For $\begin{pmatrix} \lambda_1 & b \\ 0 & \lambda_2 \end{pmatrix}$ the characteristic polynomial is $(\lambda_1 - \lambda)(\lambda_2 - \lambda) - b \cdot 0$, so the eigenvalues are the diagonal entries, and $(1, 0)^T$ is an eigenvector for λ_1. The same holds for lower-triangular matrices (with $(0, 1)^T$ for λ_2) and, with both eigenvectors along the axes, for diagonal matrices. Robinson simply treats matrices of the form $\begin{pmatrix} \lambda_1 & b \\ 0 & \lambda_2 \end{pmatrix}$ without naming them; the lecture calls them triangular, and the notes record that for diagonal and upper-triangular matrices the eigenvalues are the diagonal entries [Notes, p. 9].

The roots of (27.8) fall into three cases, each with its own chapter [p. 261]; the class notes call them Case 1, Case 2 and Case 3 (pp. 9–10):

DISCRIMINANT $(\operatorname{tr} A)^2 - 4\det A$	EIGENVALUES	EXAMPLE	TREATED IN
> 0	Real, distinct	27.1: $6, -1$	Ch. 28 – Sections 5–6
$= 0$	One repeated real	27.4: $A = \begin{pmatrix} 5 & -4 \\ 1 & 1 \end{pmatrix}$, $(\lambda - 3)^2 = 0$, $\mathbf{v} = (2,1)^T$ only	Ch. 30 – Section 8
< 0	Complex pair $\rho \pm i\omega$	27.5: $A = \begin{pmatrix} 2 & 5 \\ -2 & 0 \end{pmatrix}$, $\lambda = 1 \pm 3i$	Ch. 29 – Section 7

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5. Distinct real eigenvalues: solution and change of basis

CASE 1: TWO REAL EIGENVALUES $\lambda_1 \neq \lambda_2$ [NOTES, P. 9]

$\mathbf{x}_1(t) = e^{\lambda_1 t}\mathbf{v}_{\lambda_1}$ and $\mathbf{x}_2(t) = e^{\lambda_2 t}\mathbf{v}_{\lambda_2}$ are both solutions of $\dot{\mathbf{x}} = A\mathbf{x}$, and the general solution is

$$\mathbf{x}(t) = C_1 e^{\lambda_1 t}\mathbf{v}_{\lambda_1} + C_2 e^{\lambda_2 t}\mathbf{v}_{\lambda_2},$$

where we need to check that $\mathbf{v}_{\lambda_1}, \mathbf{v}_{\lambda_2}$ are linearly independent, and C_1, C_2 are fixed by $\mathbf{x}_0$. Therefore $\{\mathbf{x}_1, \mathbf{x}_2\}$ is a fundamental set of solutions. The next step is to draw the general solution in a phase portrait, using a change of basis.

If $\lambda_1 \neq \lambda_2$ are real with eigenvectors $\mathbf{v}_1, \mathbf{v}_2$, then $e^{\lambda_1 t}\mathbf{v}_1$ and $e^{\lambda_2 t}\mathbf{v}_2$ are solutions, and by superposition [Eq. (28.1), p. 270]

$$\mathbf{x}(t) = Ae^{\lambda_1 t}\mathbf{v}_1 + Be^{\lambda_2 t}\mathbf{v}_2. \tag{28.1}$$

This is the general solution because the two solutions are independent. Their Wronskian at $t = 0$ is $\det[\mathbf{v}_1\ \mathbf{v}_2]$, non-zero by the next result.

PROPOSITION – EIGENVECTORS OF DISTINCT EIGENVALUES ARE INDEPENDENT [ROBINSON, APPENDIX B, P. 384]

If $A\mathbf{v}_1 = \lambda_1\mathbf{v}_1$, $A\mathbf{v}_2 = \lambda_2\mathbf{v}_2$ with $\mathbf{v}_1, \mathbf{v}_2 \neq \mathbf{0}$ and $\lambda_1 \neq \lambda_2$, then $\mathbf{v}_1, \mathbf{v}_2$ are linearly independent.

PROOF

Suppose $\alpha\mathbf{v}_1 + \beta\mathbf{v}_2 = \mathbf{0}$ (B.3). Applying A gives $\alpha\lambda_1\mathbf{v}_1 + \beta\lambda_2\mathbf{v}_2 = \mathbf{0}$ (B.4). Multiply (B.3) by λ_1 and subtract (B.4):

$$\beta(\lambda_1 - \lambda_2)\mathbf{v}_2 = \mathbf{0}.$$

Since $\lambda_1 \neq \lambda_2$ and $\mathbf{v}_2 \neq \mathbf{0}$, $\beta = 0$; then (B.3) reads $\alpha\mathbf{v}_1 = \mathbf{0}$, so $\alpha = 0$. (Robinson argues by dividing by α and λ_1; subtracting as here avoids having to treat $\alpha = 0$ or $\lambda_1 = 0$ separately.)

■

EXAMPLE 28.1 (PP. 270–271), THE EXAMPLE OF THE NOTES: $\dot{x} = x + y$, $\dot{y} = 4x - 2y$ [NOTES, P. 9]

$A = \begin{pmatrix} 1 & 1 \\ 4 & -2 \end{pmatrix}$: $(1-\lambda)(-2-\lambda) - 4 = \lambda^2 + \lambda - 6 = 0, \lambda = 2, -3$. $\lambda = 2$: $(A - 2I) = \begin{pmatrix} -1 & 1 \\ 4 & -4 \end{pmatrix}$, $\mathbf{v}_1 = (1,1)^T$. $\lambda = -3$: $(A + 3I) = \begin{pmatrix} 4 & 1 \\ 4 & 1 \end{pmatrix}$, $\mathbf{v}_2 = (1,-4)^T$. Independence directly: $\alpha(1,1) + \beta(1,-4) = \mathbf{0}$ gives $\alpha + \beta = 0, \alpha - 4\beta = 0$, so $\alpha = \beta = 0$.

$$\mathbf{x}(t) = Ae^{2t}\begin{pmatrix} 1 \\ 1 \end{pmatrix} + Be^{-3t}\begin{pmatrix} 1 \\ -4 \end{pmatrix}.$$

This is the homogeneous part of Example 26.1, as it should be.

5.1 Changing coordinates: diagonalisation [Sec. 28.2, pp. 271–274]

Because $\mathbf{v}_1, \mathbf{v}_2$ are independent, every vector can be written in the eigenvector basis, $\mathbf{x} = \tilde{x}\mathbf{v}_1 + \tilde{y}\mathbf{v}_2$ (28.3). In matrix form, with $P = [\mathbf{v}_1\ \mathbf{v}_2]$ (eigenvectors as columns),

$$\mathbf{x} = P\tilde{\mathbf{x}}, \qquad \tilde{\mathbf{x}} = P^{-1}\mathbf{x}. \tag{28.4-28.5}$$

DIAGONALISATION [NOTES, P. 9]

If A has a basis of eigenvectors $\mathbf{v}_{\lambda_1}, \mathbf{v}_{\lambda_2}$, we can rewrite it as $A = PDP^{-1}$, where $P = (\mathbf{v}_{\lambda_1}, \mathbf{v}_{\lambda_2})$ and $D = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix} = \operatorname{diag}(\lambda_1, \lambda_2)$. Make the change of variable $\mathbf{z} = P^{-1}\mathbf{x}$. Then

$$\dot{\mathbf{z}} = P^{-1}\dot{\mathbf{x}} = P^{-1}A\mathbf{x} = P^{-1}\big(PDP^{-1}\big)\mathbf{x} = DP^{-1}\mathbf{x} = D\mathbf{z},$$

so $\mathbf{z} = (z_1, z_2)^T$ satisfies $\dot{z}_1 = \lambda_1 z_1$, $\dot{z}_2 = \lambda_2 z_2$, with solutions $z_1(t) = C_1 e^{\lambda_1 t}$, $z_2(t) = C_2 e^{\lambda_2 t}$. Now graph the solution (z_1, z_2) in the plane (Section 6).

(The lecture writes $\mathbf{z}$ for Robinson's $\tilde{\mathbf{x}}$.) In Robinson's words: differentiate the second relation; P^{-1} is constant:

$$\frac{d\tilde{\mathbf{x}}}{dt} = P^{-1}\frac{d\mathbf{x}}{dt} = P^{-1}A\mathbf{x} = P^{-1}AP\,\tilde{\mathbf{x}}. \tag{28.7}$$

Now compute $P^{-1}AP$ column by column, using $A[\mathbf{v}_1\ \mathbf{v}_2] = [A\mathbf{v}_1\ A\mathbf{v}_2]$:

$$AP = [\lambda_1\mathbf{v}_1\ \ \lambda_2\mathbf{v}_2] = [\mathbf{v}_1\ \mathbf{v}_2]\begin{pmatrix}\lambda_1 & 0\\ 0 & \lambda_2\end{pmatrix} = P\Lambda \quad\Longrightarrow\quad P^{-1}AP = \Lambda = \begin{pmatrix}\lambda_1 & 0\\ 0 & \lambda_2\end{pmatrix}.$$

So in the new coordinates the system is **uncoupled** [Eqs. (28.8)-(28.9)]:

$$\dot{\tilde{x}} = \lambda_1\tilde{x}, \quad \dot{\tilde{y}} = \lambda_2\tilde{y} \quad\Longrightarrow\quad \tilde{x} = Ae^{\lambda_1 t}, \quad \tilde{y} = Be^{\lambda_2 t},$$

and $\mathbf{x} = P\tilde{\mathbf{x}} = Ae^{\lambda_1 t}\mathbf{v}_1 + Be^{\lambda_2 t}\mathbf{v}_2$ recovers (28.1).

EXAMPLE 28.2 (PP. 274-276): DECOUPLING $\dot{x} = x + y,\ \dot{y} = 4x - 2y$

$P = \begin{pmatrix}1 & 1\\ 1 & -4\end{pmatrix}$, $\det P = -4 - 1 = -5 \neq 0$, and with $\begin{pmatrix}a & b\\ c & d\end{pmatrix}^{-1} = \frac{1}{ad-bc}\begin{pmatrix}d & -b\\ -c & a\end{pmatrix}$ (Appendix B), $P^{-1} = \frac{1}{5}\begin{pmatrix}4 & 1\\ 1 & -1\end{pmatrix}$, i.e. $\tilde{x} = \frac{1}{5}(4x + y)$, $\tilde{y} = \frac{1}{5}(x - y)$. Then

$$P^{-1}AP = \frac{1}{5}\begin{pmatrix}4 & 1\\ 1 & -1\end{pmatrix}\begin{pmatrix}1 & 1\\ 4 & -2\end{pmatrix}\begin{pmatrix}1 & 1\\ 1 & -4\end{pmatrix} = \frac{1}{5}\begin{pmatrix}4 & 1\\ 1 & -1\end{pmatrix}\begin{pmatrix}2 & -3\\ 2 & 12\end{pmatrix} = \frac{1}{5}\begin{pmatrix}10 & 0\\ 0 & -15\end{pmatrix} = \begin{pmatrix}2 & 0\\ 0 & -3\end{pmatrix}\ \checkmark.$$

Robinson stresses that this is not the efficient way to *solve* the system – (28.1) is – but it is the key to *drawing* it.

6. Phase portraits: nodes, saddles and a zero eigenvalue

A **phase portrait** shows the curves traced in the (x, y) plane by solutions, with arrows for increasing t. The origin is always a stationary solution of $\dot{\mathbf{x}} = A\mathbf{x}$ (and the only one when $\det A \neq 0$). Draw first the uncoupled system, then "twist" it.

6.1 The uncoupled system [Sec. 28.3, pp. 276-278]

Solutions of $\dot{\tilde{x}} = \lambda_1\tilde{x}, \dot{\tilde{y}} = \lambda_2\tilde{y}$ are $\tilde{x} = \tilde{x}_0 e^{\lambda_1 t}, \tilde{y} = \tilde{y}_0 e^{\lambda_2 t}$ (28.12). To find the curve they trace, eliminate t: $e^t = (\tilde{x}/\tilde{x}_0)^{1/\lambda_1}$, so

$$\tilde{y} = \tilde{y}_0\, e^{\lambda_2 t} = \tilde{y}_0 \left(\frac{\tilde{x}}{\tilde{x}_0}\right)^{\lambda_2/\lambda_1} = K\, \tilde{x}^{\lambda_2/\lambda_1}, \qquad K = \tilde{y}_0\, \tilde{x}_0^{-\lambda_2/\lambda_1}.$$

(Equivalently, by the chain rule $\dfrac{d\tilde{y}}{d\tilde{x}} = \dfrac{\dot{\tilde{y}}}{\dot{\tilde{x}}} = \dfrac{\lambda_2 \tilde{y}}{\lambda_1 \tilde{x}}$, a separable equation.) The axes themselves are trajectories: data on an axis stay on it.

- **Stable node** ($\lambda_1 < \lambda_2 < 0$): every solution tends to $\mathbf{0}$. Since $|\lambda_2| < |\lambda_1|$, the $\tilde{y}$ component decays more slowly, so trajectories arrive tangent to the slow eigendirection.
- **Unstable node** ($0 < \lambda_1 < \lambda_2$): the same picture with arrows reversed; solutions leave the origin tangent to the eigendirection of smaller $|\lambda|$.
- **Saddle** ($\lambda_2 < 0 < \lambda_1$): $\tilde{x}$ grows while $\tilde{y}$ decays; the curves $\tilde{x}\,\tilde{y}^{\,r} = K$ ($r = -\lambda_1/\lambda_2 > 0$) are hyperbola-like. Only solutions starting on the $\tilde{y}$-axis reach the origin.

THE ORBITS IN THE z-PLANE [NOTES, P. 9]

With $z_1 = C_1 e^{\lambda_1 t}$ and $z_2 = C_2 e^{\lambda_2 t}$, write $e^{\lambda_2 t} = \left(e^{\lambda_1 t}\right)^{\lambda_2/\lambda_1}$:

$$z_2(t) = C_2 e^{\lambda_2 t} = C_2 \left(e^{\lambda_1 t}\right)^{\lambda_2/\lambda_1} = C_2\, C_1^{-\lambda_2/\lambda_1} \left(C_1 e^{\lambda_1 t}\right)^{\lambda_2/\lambda_1} = \tilde{C}\, \big(z_1(t)\big)^{\lambda_2/\lambda_1},$$

with $\tilde{C} = C_2 C_1^{-\lambda_2/\lambda_1}$ (for $C_1 > 0$; for $C_1 < 0$ use $|z_1|$ and $|C_1|$, by symmetry). The exponent $r = \lambda_2/\lambda_1$ decides the shape.

ORDERING (NOTES)	$r = \lambda_2/\lambda_1$	ORIGIN	ORBITS NEAR 0
$0 < \lambda_1 < \lambda_2$	$r > 1$	unstable node	leave tangent to the z_1-axis, like $z_2 = \tilde{C} z_1^{\,r}$ (flat at 0)
$0 < \lambda_2 < \lambda_1$	$0 < r < 1$	unstable node	leave tangent to the z_2-axis
$\lambda_2 < \lambda_1 < 0$	$r > 1$	stable node	enter tangent to the z_1-axis
$\lambda_1 < \lambda_2 < 0$	$0 < r < 1$	stable node	enter tangent to the z_2-axis
$\lambda_1 < 0 < \lambda_2$	$r < 0$	saddle	z_1-axis in, z_2-axis out; hyperbola-like curves $z_2 = \tilde{C}\lvert z_1\rvert^{-\lvert r\rvert}$
$\lambda_2 < 0 < \lambda_1$	$r < 0$	saddle	z_1-axis out, z_2-axis in

In words: at a node the orbits are tangent, at the origin, to the axis of the eigenvalue with the **smaller** modulus (the slow direction), and far away they become parallel to the other axis. (For $r > 1$, $dz_2/dz_1 = r\tilde{C} z_1^{\,r-1} \to 0$ as $z_1 \to 0$; for $0 < r < 1$ it tends to infinity.)

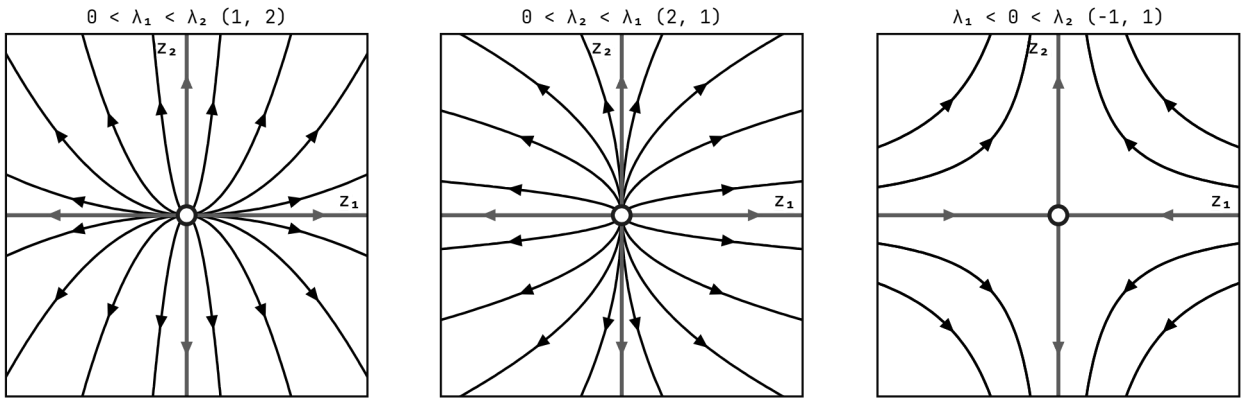


Figure 4.1 (a). *The uncoupled system $\dot z_1 = \lambda_1 z_1$, $\dot z_2 = \lambda_2 z_2$ in the $\mathbf{z}$-plane, with the orderings drawn in the class notes (p. 9). Amber: the axes, which are orbits. Unstable nodes leave the origin tangent to the axis of the eigenvalue with the smaller modulus; at a saddle the axis of the negative eigenvalue comes in and the other goes out.*

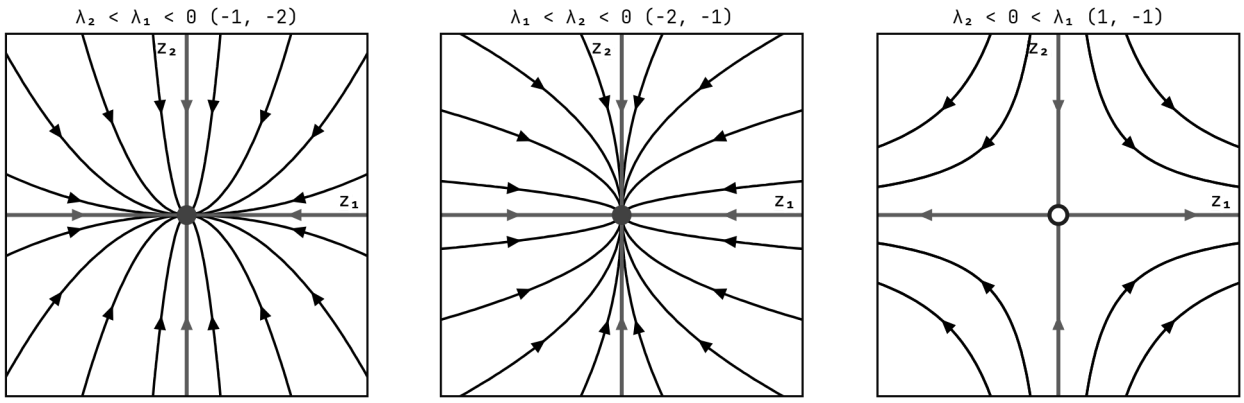


Figure 4.1 (b). *The same with the arrows reversed: stable nodes, entering the origin tangent to the axis of the eigenvalue with the smaller modulus, and the other saddle. Each orbit is the curve $z_2 = \tilde C\, z_1^{\lambda_2/\lambda_1}$.*

6.2 Back to the original coordinates [Sec. 28.4, p. 279]

The map $\mathbf{x} = P\tilde{\mathbf{x}}$ sends the $\tilde x$-axis to the line through $\mathbf{v}_1$ and the $\tilde y$-axis to the line through $\mathbf{v}_2$. Straight lines stay straight and the origin stays fixed, so the picture is the uncoupled one, sheared and rotated. Practical recipe:

BACK TO x [NOTES, P. 9]

$$\mathbf{x} = P\mathbf{z} = (\mathbf{v}_{\lambda_1}, \mathbf{v}_{\lambda_2}) \begin{pmatrix} z_1 \\ z_2 \end{pmatrix} = C_1 e^{\lambda_1 t}\mathbf{v}_{\lambda_1} + C_2 e^{\lambda_2 t}\mathbf{v}_{\lambda_2}.$$

Example of the notes. Find the solution and draw the phase diagram of $\dot x = x + y$, $\dot y = 4x - 2y$, i.e. $A = \begin{pmatrix} 1 & 1 \\ 4 & -2 \end{pmatrix}$. Eigenvalues: $\lambda^2 + \lambda - 6 = (\lambda - 2)(\lambda + 3) = 0$, $\lambda_1 = 2$, $\lambda_2 = -3$. Eigenvectors: $\mathbf{v}_1 = (1, 1)^T$ and $\mathbf{v}_2 = (1, -4)^T$ (Example 28.1). General solution $\mathbf{x}(t) = C_1 e^{2t}(1, 1)^T + C_2 e^{-3t}(1, -4)^T$.

Phase diagram: in $\mathbf{z} = P^{-1}\mathbf{x}$, since $\lambda_2 < 0 < \lambda_1$, a saddle (last panel of Figure 4.1 (b)); to go back to $\mathbf{x}$, $\mathbf{x} = P\mathbf{z} = (\mathbf{v}_1, \mathbf{v}_2)\mathbf{z}$ sends the z_1-axis to the line of $\mathbf{v}_1$ (out) and the z_2-axis to the line of $\mathbf{v}_2$ (in): Figure 4.2.

1. Find the eigenvalues and eigenvectors.
2. Draw the eigenvector lines; put arrows pointing away from the origin if $\lambda > 0$ and towards it if $\lambda < 0$.
3. Fill in the other trajectories consistently with the type (node or saddle), tangent to the slow direction near the origin and parallel to the fast one far away.

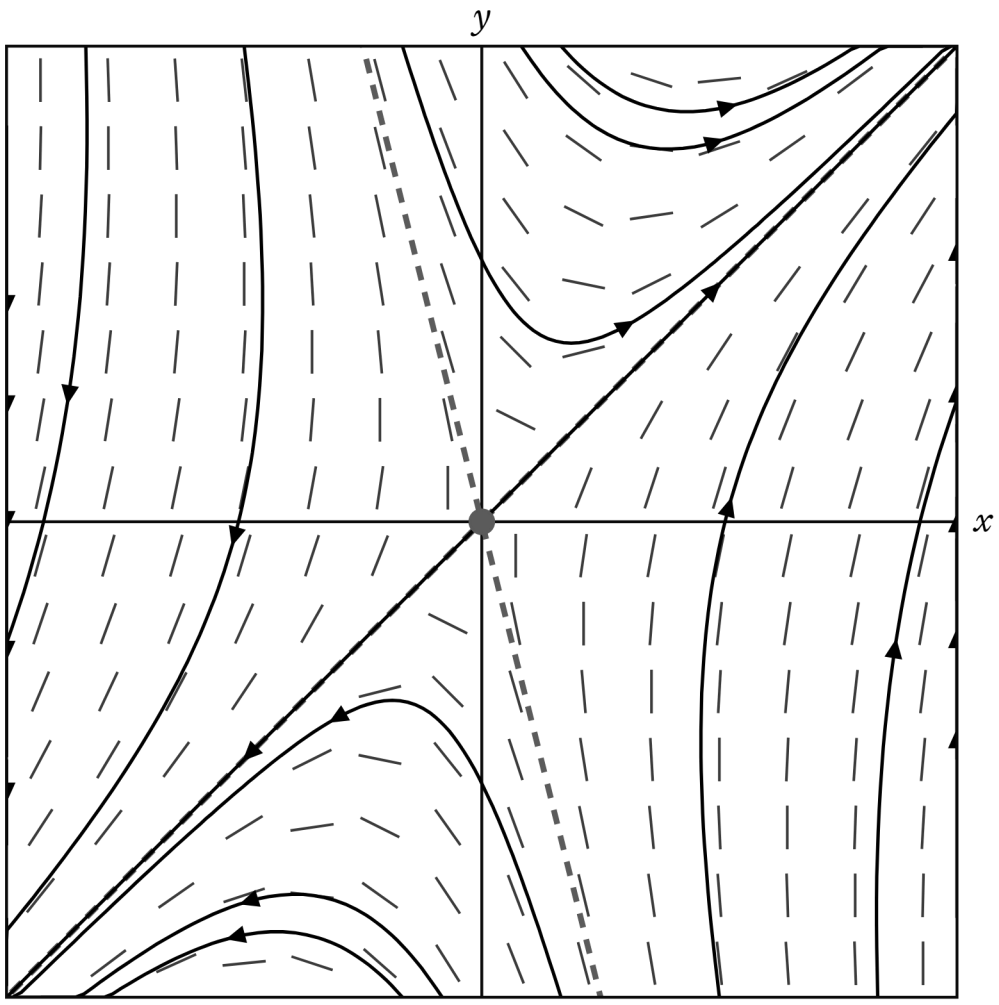


Figure 4.2. *Phase portrait of $\dot{x} = x + y$, $\dot{y} = 4x - 2y$ (Robinson Example 28.3, Fig. 28.8). Dashed amber: the eigenvector lines $y = x$ ($\lambda_1 = 2$, outward) and $y = -4x$ ($\lambda_2 = -3$, inward). The origin is a saddle; the line $y = -4x$ is its stable manifold and $y = x$ its unstable manifold.*

6.3 Stable and unstable manifolds [Sec. 28.5, pp. 281–282]

DEFINITIONS

The **stable manifold** $W^s(\mathbf{0})$ is the set of initial points whose trajectories tend to the origin as $t \to +\infty$; the **unstable manifold** $W^u(\mathbf{0})$ is the set whose trajectories tend to the origin as $t \to -\infty$.

	$W^s(\mathbf{0})$	$W^u(\mathbf{0})$
Stable node	the whole plane	$\{\mathbf{0}\}$
Saddle	the eigenvector line of the negative eigenvalue	the eigenvector line of the positive eigenvalue
Unstable node	$\{\mathbf{0}\}$	the whole plane

At a saddle the stable manifold is a **separatrix**: points on one side are carried to infinity along one branch of W^u, points on the other side along the opposite branch. In Example 28.3, initial data above the line $y = -4x$ end up with $x, y \to +\infty$; data below it with $x, y \to -\infty$.

TERMINOLOGY

The lecture notes mention a "saddle node"; in the phase-portrait chapters Robinson says only "saddle point" or "saddle" and reserves "node" for stable and unstable nodes, so the intended term is almost certainly "saddle". (In bifurcation theory, Topic 2, a *saddle-node* is something else.)

6.4 Special case: a zero eigenvalue [Notes, p. 9]

The notes end Case 1 with a special case: one eigenvalue is 0. Say $\lambda_1 = 0 \neq \lambda_2$. Then $A\mathbf{v}_1 = \mathbf{0}$, so every point $c\,\mathbf{v}_1$ of the line span $\mathbf{v}_1$ is an **equilibrium** ($\mathbf{x}(t) \equiv c\mathbf{v}_1$ solves $\dot{\mathbf{x}} = A\mathbf{x}$, since both sides vanish). With $e^{0t} = 1$ the general solution of Case 1 becomes

$$\mathbf{x}(t) = C_1\mathbf{v}_1 + C_2 e^{\lambda_2 t}\mathbf{v}_2.$$

- Each orbit is a straight half-line parallel to $\mathbf{v}_2$, through the point $C_1\mathbf{v}_1$ of the line of equilibria.
- $\lambda_2 < 0$: every solution tends to $C_1\mathbf{v}_1$; $\lambda_2 > 0$: every solution leaves the line. The origin is no longer an isolated equilibrium ($\det A = \lambda_1\lambda_2 = 0$).
- In the $\mathbf{z}$-plane: $z_1 = C_1$ is constant and $z_2 = C_2 e^{\lambda_2 t}$, so the orbits are vertical lines and the z_1-axis is made of equilibria.

EXAMPLE (WRITTEN FOR THIS PAGE): $A = \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix}$

1. $\det(A - \lambda I) = (-1-\lambda)^2 - 1 = \lambda^2 + 2\lambda = \lambda(\lambda+2) = 0$: $\lambda_1 = 0, \lambda_2 = -2$.
2. $\lambda_1 = 0$: $-a + b = 0$, $\mathbf{v}_1 = (1,1)^T$. $\lambda_2 = -2$: $(A + 2I)\mathbf{v} = \mathbf{0}$ gives $a + b = 0$, $\mathbf{v}_2 = (1,-1)^T$.
3. $\mathbf{x}(t) = C_1(1,1)^T + C_2 e^{-2t}(1,-1)^T$. From $\mathbf{x}(0) = (x_{10}, x_{20})^T$: $C_1 + C_2 = x_{10}$, $C_1 - C_2 = x_{20}$, so $C_1 = \tfrac{1}{2}(x_{10} + x_{20})$, $C_2 = \tfrac{1}{2}(x_{10} - x_{20})$.
4. As $t \to \infty$, $\mathbf{x}(t) \to C_1(1,1)^T$: the solution slides along the direction $(1,-1)^T$ onto the diagonal, keeping $x_1 + x_2$ constant (indeed $\dot{x}_1 + \dot{x}_2 = 0$).

Check: $A(1,1)^T = (0,0)^T$ and $A(1,-1)^T = (-2,2)^T = -2(1,-1)^T$. Exercise 11 of the Chapter 4 (part 1) sheet treats this case with other matrices.

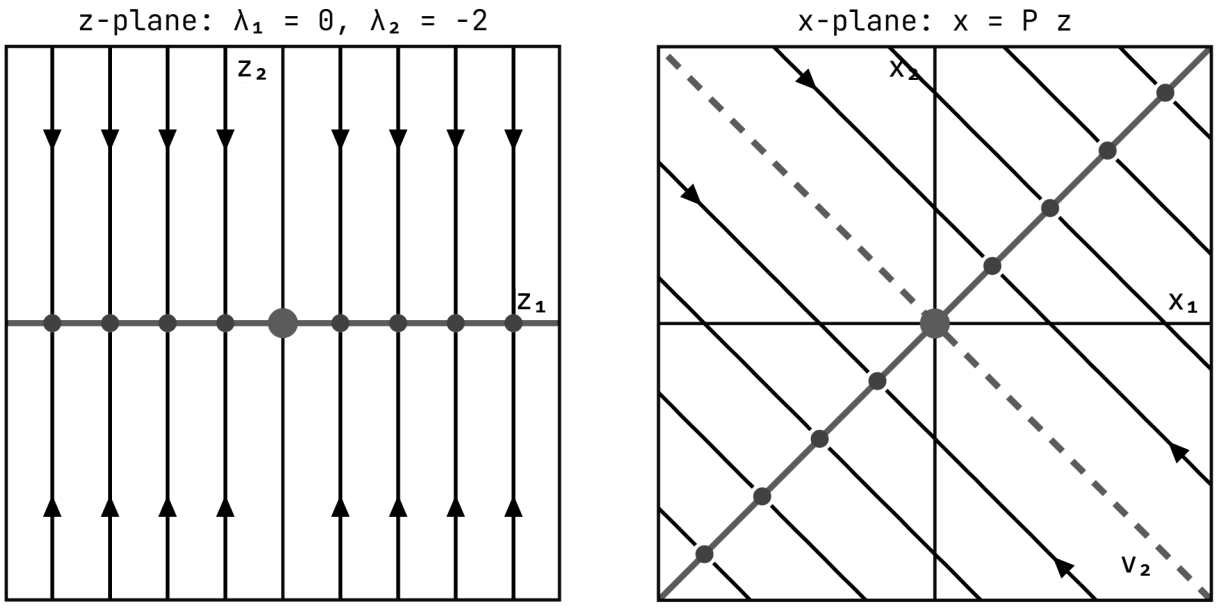


Figure 4.3. *A zero eigenvalue: $\dot{\mathbf{x}} = A\mathbf{x}$ with $A = \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix}$, $\lambda_1 = 0$ with $\mathbf{v}_1 = (1,1)^T$ and $\lambda_2 = -2$ with $\mathbf{v}_2 = (1,-1)^T$. Amber: the line $\operatorname{span}\mathbf{v}_1$, made of equilibria (dots). Every other orbit is a straight segment parallel to $\mathbf{v}_2$ that ends on that line, as in the sketch of the class notes.*

7. Complex eigenvalues: spirals and centres

Let A be a real 2×2 matrix whose characteristic polynomial has a negative discriminant, $\Delta = (\operatorname{tr} A)^2 - 4 \det A < 0$. Its eigenvalues are then a complex-conjugate pair

$$\lambda_\pm = \rho \pm i\omega, \qquad \rho = \tfrac{1}{2}\operatorname{tr} A, \quad \omega = \tfrac{1}{2}\sqrt{-\Delta} > 0, \tag{7.1}$$

and the eigenvector of λ_- is the complex conjugate of the eigenvector of λ_+ (because A is real: conjugate $A\mathbf{v} = \lambda\mathbf{v}$ to get $A\bar{\mathbf{v}} = \bar{\lambda}\bar{\mathbf{v}}$). Write the eigenvector of λ_+ through its real and imaginary parts, $\mathbf{v} = \mathbf{u} + i\mathbf{w}$ with $\mathbf{u}, \mathbf{w} \in \mathbb{R}^2$.

CASE 2: COMPLEX CONJUGATE EIGENVALUES [NOTES, P. 10]

Suppose A has an eigenvalue $\lambda = \rho \pm i\omega$. Then it has eigenvectors $\mathbf{v}_\pm = \mathbf{v}_1 \pm i\,\mathbf{v}_2$ (linear algebra). The notes reach the canonical form through a complex diagonalisation (Section 7.5) and give

$$\text{general solution:} \quad \mathbf{x}(t) = C_1 \operatorname{Re}\left\{\mathbf{v}e^{\lambda t}\right\} + C_2 \operatorname{Im}\left\{\mathbf{v}e^{\lambda t}\right\},$$

$$\text{particular solution:} \quad \mathbf{x}(t) = r(0)\, e^{\rho t}\Big(\cos\big(\theta(0) - \omega t\big)\,\mathbf{v}_1 + \sin\big(\theta(0) - \omega t\big)\,\mathbf{v}_2\Big).$$

Both are derived below (Sections 7.2 and 7.8).

NOTATION OF THIS SECTION

The class notes and Robinson both write $\mathbf{v}_1 + i\mathbf{v}_2$ for the eigenvector; Robinson writes P for the change-of-basis matrix and $\tilde{\mathbf{x}}$ for the new variable, while the notes use Q and $\mathbf{z}$, and so does this section: $\mathbf{u} \equiv \mathbf{v}_1$, $\mathbf{w} \equiv \mathbf{v}_2$, $Q \equiv P$ (Robinson) and $\mathbf{z} \equiv \tilde{\mathbf{x}}$. Section 5 called the matrix of eigenvectors P; from here on every change of basis is called Q.

7.1 A guided example: what to expect before the theory

[Examples 27.5 and 29.1, pp. 265–266 and 286]

EXAMPLE 7.1 – $\dot{x} = 2x + 5y$, $\dot{y} = -2x$ (FIRST PASS: TYPE AND SENSE OF ROTATION)

Here $A = \begin{pmatrix} 2 & 5 \\ -2 & 0 \end{pmatrix}$, $\operatorname{tr} A = 2$, $\det A = 2 \cdot 0 - 5 \cdot (-2) = 10$.

1. **Eigenvalues.** $\det(A - \lambda I) = (2-\lambda)(0-\lambda) - (5)(-2) = \lambda^2 - 2\lambda + 10 = 0$, so $\lambda = \dfrac{2 \pm \sqrt{4 - 40}}{2} = 1 \pm 3i$. Hence $\rho = 1, \omega = 3$ and $\Delta = -36 < 0$.
2. **Type.** $\rho = 1 > 0$: an *unstable spiral* (radius grows like e^{t}).
3. **Eigenvector for $\lambda = 1 + 3i$.** The first row of $(A - \lambda I)\mathbf{v} = \mathbf{0}$ is $(2 - 1 - 3i)v_1 + 5v_2 = (1 - 3i)v_1 + 5v_2 = 0$. Take $v_1 = 5$; then $v_2 = -(1 - 3i) = -1 + 3i$, and $\mathbf{v} = \begin{pmatrix} 5 \\ -1 + 3i \end{pmatrix} = \underbrace{\begin{pmatrix} 5 \\ -1 \end{pmatrix}}_{\mathbf{u}} + i\underbrace{\begin{pmatrix} 0 \\ 3 \end{pmatrix}}_{\mathbf{w}}$. Check with the second row: $-2 \cdot 5 + (0 - \lambda)(-1 + 3i) = -10 - (1 + 3i)(-1 + 3i) = -10 - (-1 - 9) = 0$.

4. **Sense of rotation, without any change of basis.** Evaluate the vector field at $(0,1)^T$ (top of the plane): $\dot{\mathbf{x}} = A(0,1)^T = (5,0)^T$, which points to the right. At the top, motion to the right is *clockwise*. Confirm at $(1,0)^T$: $\dot{\mathbf{x}} = (2,-2)^T$ points down on the right-hand side, again clockwise.

5. **Sketch.** A spiral that turns clockwise and moves outward, drawn as a distorted spiral because it lives in the (x,y) basis (Fig. 4.5, right). The rest of this section explains *why* it is distorted exactly as it is, and gives its equation.

7.2 The real general solution [Sec. 29.1, pp. 285–286]

As for distinct real eigenvalues, $e^{\lambda_+ t}\mathbf{v}$ and $e^{\lambda_- t}\bar{\mathbf{v}}$ solve $\dot{\mathbf{x}} = A\mathbf{x}$, and so does every combination $Ce^{\lambda_+ t}\mathbf{v} + De^{\lambda_- t}\bar{\mathbf{v}}$ with complex C, D. To get a *real* solution choose $D = \bar{C}$: the second term is then the conjugate of the first and the sum is twice a real part, because $z + \bar{z} = 2\,\mathrm{Re}\,z$:

$$\mathbf{x}(t) = 2\,\mathrm{Re}\left[Ce^{(\rho+i\omega)t}(\mathbf{u}+i\mathbf{w})\right]. \tag{7.2}$$

Write $C = \alpha + i\beta$ and expand with Euler's formula $e^{i\omega t} = \cos\omega t + i\sin\omega t$:

1. $Ce^{i\omega t} = (\alpha+i\beta)(\cos\omega t + i\sin\omega t) = (\alpha\cos\omega t - \beta\sin\omega t) + i(\beta\cos\omega t + \alpha\sin\omega t)$.
2. Multiply by $\mathbf{u}+i\mathbf{w}$ and keep only the real part ($i\cdot i = -1$): $\mathrm{Re}\,[(a+ib)(\mathbf{u}+i\mathbf{w})] = a\mathbf{u} - b\mathbf{w}$, with $a = \alpha\cos\omega t - \beta\sin\omega t$ and $b = \beta\cos\omega t + \alpha\sin\omega t$.
3. Therefore $\mathbf{x}(t) = 2e^{\rho t}\big[(\alpha\cos\omega t - \beta\sin\omega t)\mathbf{u} - (\beta\cos\omega t + \alpha\sin\omega t)\mathbf{w}\big]$.
4. Put $a_1 = 2\alpha$ and $b_1 = -2\beta$ (two arbitrary real constants):

$$\mathbf{x}(t) = e^{\rho t}\Big[\big(a_1\cos\omega t + b_1\sin\omega t\big)\mathbf{u} + \big(b_1\cos\omega t - a_1\sin\omega t\big)\mathbf{w}\Big]. \tag{7.3}$$

This is Robinson's (29.2) with $\mathbf{v}_1 = \mathbf{u}$, $\mathbf{v}_2 = \mathbf{w}$. It already contains everything, but it does not show *how* the solution moves. The change of basis does.

The form of the notes. The real and imaginary parts of the single complex solution $\mathbf{v}e^{\lambda t}$ are themselves real solutions (the real and imaginary parts of $\dot{\mathbf{y}} - A\mathbf{y} = \mathbf{0}$ vanish separately because A is real). By Euler's formula,

$$\mathbf{v}e^{\lambda t} = e^{\rho t}(\mathbf{u}+i\mathbf{w})(\cos\omega t + i\sin\omega t) = e^{\rho t}\Big[\big(\cos\omega t\,\mathbf{u} - \sin\omega t\,\mathbf{w}\big) + i\big(\sin\omega t\,\mathbf{u} + \cos\omega t\,\mathbf{w}\big)\Big],$$

so

$$C_1\,\mathrm{Re}\left\{\mathbf{v}e^{\lambda t}\right\} + C_2\,\mathrm{Im}\left\{\mathbf{v}e^{\lambda t}\right\} = e^{\rho t}\Big[\big(C_1\cos\omega t + C_2\sin\omega t\big)\mathbf{u} + \big(C_2\cos\omega t - C_1\sin\omega t\big)\mathbf{w}\Big],$$

which is (7.3) with $a_1 = C_1$ and $b_1 = C_2$ [Notes, p. 10].

7.3 How A acts on $\mathbf{u}$ and $\mathbf{w}$ [Sec. 29.2, p. 287, Eq. (29.3)]

Separate the real and imaginary parts of $A\mathbf{v} = \lambda\mathbf{v}$. The left side is $A(\mathbf{u} + i\mathbf{w}) = A\mathbf{u} + iA\mathbf{w}$ (both real vectors). The right side is

$$(\rho + i\omega)(\mathbf{u} + i\mathbf{w}) = \rho\mathbf{u} + i\rho\mathbf{w} + i\omega\mathbf{u} + i^2\omega\mathbf{w} = (\rho\mathbf{u} - \omega\mathbf{w}) + i(\omega\mathbf{u} + \rho\mathbf{w}).$$

Two complex vectors are equal exactly when their real parts and their imaginary parts are equal:

$$\boxed{A\mathbf{u} = \rho\mathbf{u} - \omega\mathbf{w}, \qquad A\mathbf{w} = \omega\mathbf{u} + \rho\mathbf{w}.} \tag{7.4}$$

So A maps the plane $\mathrm{span}\{\mathbf{u}, \mathbf{w}\}$ into itself: it scales by ρ and mixes $\mathbf{u}$ with $\mathbf{w}$ through the rotation angle ω. There is no real eigenvector, but there is an invariant *basis*.

LEMMA — $\mathbf{u}$ AND $\mathbf{w}$ ARE LINEARLY INDEPENDENT [EXERCISE 29.4]
If $\omega \neq 0$, the real vectors $\mathbf{u}, \mathbf{w}$ form a basis of $\mathbb{R}^2$.

PROOF
Suppose they were dependent. Then the eigenvector $\mathbf{v} = \mathbf{u} + i\mathbf{w}$ would be a complex multiple of one real vector $\mathbf{r} \neq \mathbf{0}$ (if $\mathbf{w} = c\mathbf{u}$ then $\mathbf{v} = (1 + ic)\mathbf{u}$; if $\mathbf{u} = \mathbf{0}$ then $\mathbf{v} = i\mathbf{w}$): $\mathbf{v} = \kappa\mathbf{r}$ with $\kappa \in \mathbb{C} \setminus \{0\}$. Then $A\mathbf{v} = \kappa A\mathbf{r}$ and $\lambda\mathbf{v} = \lambda\kappa\mathbf{r}$, so $A\mathbf{r} = \lambda\mathbf{r}$. The left side is a real vector, so $\lambda\mathbf{r}$ is real and, as $\mathbf{r} \neq \mathbf{0}$ is real, λ must be real. This contradicts $\mathrm{Im}\,\lambda = \omega \neq 0$.

■

7.4 The change of basis $\mathbf{z} = Q^{-1}\mathbf{x}$ [Sec. 29.2, pp. 287–288, Eqs. (29.4)–(29.7)]

Because $\mathbf{u}, \mathbf{w}$ are a basis, every $\mathbf{x}$ has unique coordinates (z_1, z_2): $\mathbf{x} = z_1\mathbf{u} + z_2\mathbf{w}$. In matrix form, with the columns **in the order $\mathbf{u}$, then $\mathbf{w}$**,

$$Q = [\,\mathbf{u} \mid \mathbf{w}\,], \qquad \mathbf{x} = Q\mathbf{z}, \qquad \mathbf{z} = Q^{-1}\mathbf{x}. \tag{7.5}$$

Q is constant, so differentiating $\mathbf{z} = Q^{-1}\mathbf{x}$ gives $\dot{\mathbf{z}} = Q^{-1}\dot{\mathbf{x}} = Q^{-1}A\mathbf{x} = Q^{-1}AQ\,\mathbf{z}$. Compute $Q^{-1}AQ$ column by column, using $AQ = [A\mathbf{u} \mid A\mathbf{w}]$ and (7.4):

$$AQ = [\rho\mathbf{u} - \omega\mathbf{w} \mid \omega\mathbf{u} + \rho\mathbf{w}] = [\,\mathbf{u} \mid \mathbf{w}\,] \begin{pmatrix} \rho & \omega \\ -\omega & \rho \end{pmatrix} = Q\,J.$$

The last equality is the rule "QM has as its j-th column the combination of the columns of Q with the coefficients of the j-th column of M": the first column of J is $(\rho, -\omega)^T$, giving $\rho\mathbf{u} - \omega\mathbf{w}$, and the second is $(\omega, \rho)^T$, giving $\omega\mathbf{u} + \rho\mathbf{w}$. Multiplying on the left by Q^{-1}:

$$\boxed{Q^{-1}AQ = J = \begin{pmatrix} \rho & \omega \\ -\omega & \rho \end{pmatrix}, \qquad \dot{z}_1 = \rho z_1 + \omega z_2, \quad \dot{z}_2 = -\omega z_1 + \rho z_2.} \tag{7.6}$$

CONVENTION CHECK: THE ORDER OF THE COLUMNS

With $Q = [\mathbf{u} \mid \mathbf{w}]$ (Robinson's choice) the canonical matrix is $J = \begin{pmatrix} \rho & \omega \\ -\omega & \rho \end{pmatrix}$ and the rotation in the $\mathbf{z}$-plane is **clockwise**. With the opposite order $Q' = [\mathbf{w} \mid \mathbf{u}]$ the same computation gives $J' = \begin{pmatrix} \rho & -\omega \\ \omega & \rho \end{pmatrix} = J^T$ and an anticlockwise rotation. Both are correct; this page always uses $[\mathbf{u} \mid \mathbf{w}]$, so the $\mathbf{z}$-plane always turns clockwise.

7.5 The route of the notes: a complex diagonalisation [Notes, p. 10]

The notes reach the same matrix J from the complex eigenvectors. With $\mathbf{v}_\pm = \mathbf{u} \pm i\mathbf{w}$ and $\lambda_\pm = \rho \pm i\omega$, A has the complex diagonalisation

$$A = PDP^{-1}, \qquad P = (\mathbf{v}_+ \mid \mathbf{v}_-), \qquad D = \operatorname{diag}(\lambda_+, \lambda_-) = \begin{pmatrix} \rho + i\omega & 0 \\ 0 & \rho - i\omega \end{pmatrix}.$$

1. **Real vectors from the complex ones.** Adding and subtracting $\mathbf{v}_\pm = \mathbf{u} \pm i\mathbf{w}$: $\mathbf{u} = \dfrac{\mathbf{v}_+ + \mathbf{v}_-}{2}$ and $\mathbf{w} = -i\,\dfrac{\mathbf{v}_+ - \mathbf{v}_-}{2}$ (check: $-i \cdot \frac{2i\mathbf{w}}{2} = \mathbf{w}$).

2. **The matrix Q in terms of P.** Each column of $Q = (\mathbf{u} \mid \mathbf{w})$ is a combination of the columns of P:

$$Q = P\,M, \qquad M = \begin{pmatrix} 1/2 & -i/2 \\ 1/2 & i/2 \end{pmatrix},$$

   since $P\,(1/2, 1/2)^T = \frac{1}{2}(\mathbf{v}_+ + \mathbf{v}_-) = \mathbf{u}$ and $P\,(-i/2, i/2)^T = -\frac{i}{2}(\mathbf{v}_+ - \mathbf{v}_-) = \mathbf{w}$.

3. **Its inverse.** $\det M = \frac{1}{2}\cdot\frac{i}{2} - \left(-\frac{i}{2}\right)\frac{1}{2} = \frac{i}{2}$, so $M^{-1} = \dfrac{1}{i/2}\begin{pmatrix} i/2 & i/2 \\ -1/2 & 1/2 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ i & -i \end{pmatrix}$, and $P = Q\begin{pmatrix} 1 & 1 \\ i & -i \end{pmatrix}$, as in the notes.

4 **The new variable.** With $\mathbf{z} = Q^{-1}\mathbf{x}$, $\dot{\mathbf{z}} = Q^{-1}A\mathbf{x} = Q^{-1}PDP^{-1}Q\,\mathbf{z} = M^{-1}DM\,\mathbf{z}$, because $Q^{-1}P = M^{-1}$ and $P^{-1}Q = M$.

5 **Multiply.** $DM = \begin{pmatrix} \frac{\rho+i\omega}{2} & -i\,\frac{\rho+i\omega}{2} \\ \frac{\rho-i\omega}{2} & i\,\frac{\rho-i\omega}{2} \end{pmatrix}$, and then

$$M^{-1}DM = \frac{1}{2}\begin{pmatrix} (\rho+i\omega)+(\rho-i\omega) & -i(\rho+i\omega)+i(\rho-i\omega) \\ i(\rho+i\omega)-i(\rho-i\omega) & (\rho+i\omega)+(\rho-i\omega) \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 2\rho & 2\omega \\ -2\omega & 2\rho \end{pmatrix} = \begin{pmatrix} \rho & \omega \\ -\omega & \rho \end{pmatrix}.$$

(Top right: $-i\rho+\omega+i\rho+\omega = 2\omega$; bottom left: $i\rho-\omega-i\rho-\omega = -2\omega$.)

This is J of (7.6): the complex route and the real route of 7.4 give the same real canonical system. The notes then solve it in polar coordinates, $z_1 = r\cos\theta$, $z_2 = r\sin\theta$, $r^2 = z_1^2+z_2^2$, $\theta = \arctan(z_2/z_1)$ [Notes, p. 10].

7.6 Solving $\dot{\mathbf{z}} = J\mathbf{z}$ in polar coordinates [Sec. 29.2, pp. 288–290]

The two equations (7.6) are coupled, but they say "rotate and scale", which polar coordinates $z_1 = r\cos\theta$, $z_2 = r\sin\theta$ express directly.

1 **The radius.** $r^2 = z_1^2+z_2^2$. Differentiate with the chain rule: $2r\dot r = 2z_1\dot z_1 + 2z_2\dot z_2$. Substitute (7.6): $2z_1(\rho z_1+\omega z_2)+2z_2(-\omega z_1+\rho z_2) = 2\rho z_1^2+2\omega z_1z_2-2\omega z_1z_2+2\rho z_2^2 = 2\rho r^2$. Hence $\dot r = \rho r$.

2 **The angle.** $\theta = \arctan(z_2/z_1)$. By the chain rule, with $\dfrac{d}{ds}\arctan s = \dfrac{1}{1+s^2}$, and the quotient rule for $s = z_2/z_1$:

$$\dot\theta = \frac{1}{1+(z_2/z_1)^2}\cdot\frac{z_1\dot z_2 - z_2\dot z_1}{z_1^2} = \frac{z_1\dot z_2 - z_2\dot z_1}{z_1^2+z_2^2}.$$

Substitute (7.6): $z_1\dot z_2 - z_2\dot z_1 = z_1(-\omega z_1+\rho z_2) - z_2(\rho z_1+\omega z_2) = -\omega z_1^2-\omega z_2^2 = -\omega r^2$. Hence $\dot\theta = -\omega$.

3 **Integrate** the two decoupled equations $\dot r = \rho r$ and $\dot\theta = -\omega$. For the first, $\dfrac{d}{dt}\left(e^{-\rho t}r\right) = e^{-\rho t}(\dot r - \rho r) = 0$, so $e^{-\rho t}r$ is constant, and Barrow's rule between 0 and t gives $e^{-\rho t}r(t) - r(0) = 0$. For the second, $\theta(t)-\theta(0) = \int_0^t(-\omega)\,ds = -\omega t$.

$$r(t) = r_0e^{\rho t}, \qquad \theta(t) = \theta_0 - \omega t.$$

4 **Back to Cartesian z.** With $z_{10} = r_0\cos\theta_0$, $z_{20} = r_0\sin\theta_0$ and the subtraction formulas $\cos(\theta_0-\omega t) = \cos\theta_0\cos\omega t + \sin\theta_0\sin\omega t$, $\sin(\theta_0-\omega t) = \sin\theta_0\cos\omega t - \cos\theta_0\sin\omega t$:

$$\mathbf{z}(t) = e^{\rho t}\begin{pmatrix} \cos\omega t & \sin\omega t \\ -\sin\omega t & \cos\omega t \end{pmatrix}\begin{pmatrix} z_{10} \\ z_{20} \end{pmatrix}. \tag{7.7}$$

5 **Check by substitution.** $\dot z_1 = \rho z_1 + e^{\rho t}\omega(-z_{10}\sin\omega t + z_{20}\cos\omega t) = \rho z_1 + \omega z_2$, and likewise $\dot z_2 = -\omega z_1 + \rho z_2$.

A shortcut worth knowing: the complex variable $\zeta = z_1 + iz_2$ satisfies $\dot\zeta = (\rho - i\omega)\zeta$, so $\zeta = \zeta_0 e^{(\rho - i\omega)t}$, whose real and imaginary parts are exactly (7.7).

7.7 Phase portrait in the z -plane: spirals and centre [Sec. 29.2, pp. 289–291]

Eliminate t using $t = (\theta_0 - \theta)/\omega$: the orbits are the *logarithmic spirals* $r = r_0 \exp\big(\rho(\theta_0 - \theta)/\omega\big)$. The motion is always clockwise because $\dot\theta = -\omega < 0$; the sign of ρ decides the radius:

SIGN OF ρ	RADIUS $r(t) = r_0 e^{\rho t}$	ORIGIN	STABILITY
$\rho < 0$	$\to 0$ as $t \to +\infty$ (and $\to \infty$ as $t \to -\infty$)	**stable spiral**	asymptotically stable
$\rho > 0$	$\to \infty$ as $t \to +\infty$ ($\to 0$ as $t \to -\infty$)	**unstable spiral**	unstable
$\rho = 0$	constant, $r(t) = r_0$	**centre**: closed circles, period $T = 2\pi/\omega$	stable, not attracting

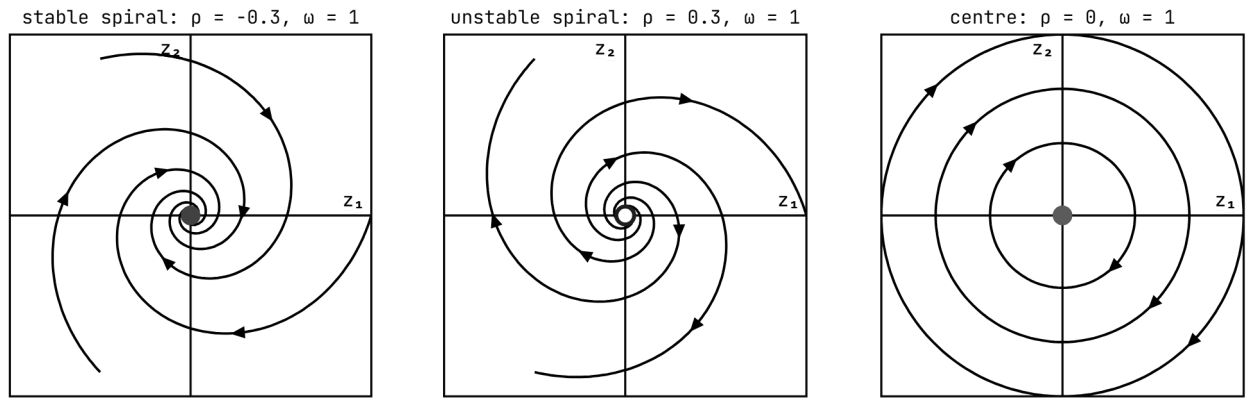


Figure 4.4. *The canonical complex system in the* $\mathbf{z}$*-plane. All three turn clockwise ($\dot\theta = -\omega$); the sign of ρ decides whether the radius shrinks, grows or stays fixed.*

The solution of a spiral keeps oscillating: each component is an exponentially modulated sine/cosine, and one full turn takes $2\pi/\omega$ whatever ρ is. A larger ratio $|\rho|/\omega$ gives a faster-collapsing (or exploding) spiral per turn: the radius changes by the factor $e^{2\pi\rho/\omega}$ in one revolution.

7.8 Back to $\mathbf{x} = Q\mathbf{z}$ [Sec. 29.3, pp. 291–293]

Write $\mathbf{x} = Q\mathbf{z} = z_1\mathbf{u} + z_2\mathbf{w}$ and insert (7.7):

$$\mathbf{x}(t) = e^{\rho t}\Big[\big(z_{10}\cos\omega t + z_{20}\sin\omega t\big)\mathbf{u} + \big(z_{20}\cos\omega t - z_{10}\sin\omega t\big)\mathbf{w}\Big]. \tag{7.8}$$

This is (7.3) with $a_1 = z_{10}$ and $b_1 = z_{20}$: the arbitrary constants are the initial coordinates, $(z_{10}, z_{20})^T = Q^{-1}\mathbf{x}_0$. In components, if $\mathbf{u} = (u_1, u_2)^T$ and $\mathbf{w} = (w_1, w_2)^T$, $x_1 = z_1u_1 + z_2w_1$ and $x_2 = z_1u_2 + z_2w_2$: each coordinate of $\mathbf{x}$ is a sum of $e^{\rho t}\cos\omega t$ and $e^{\rho t}\sin\omega t$.

The particular solution of the notes. In polar form $\mathbf{z} = r(\cos\theta, \sin\theta)^T$ with $r = r(0)e^{\rho t}$ and $\theta = \theta(0) - \omega t$, so $\mathbf{x} = z_1\mathbf{u} + z_2\mathbf{w}$ reads [Notes, p. 10]

$$\mathbf{x}(t) = r(0)\, e^{\rho t}\Big(\cos\big(\theta(0) - \omega t\big)\,\mathbf{v}_1 + \sin\big(\theta(0) - \omega t\big)\,\mathbf{v}_2\Big), \qquad \mathbf{v}_1 = \mathbf{u},\ \mathbf{v}_2 = \mathbf{w}.$$

Geometry. Q is a linear map: it sends the origin to the origin and straight lines to straight lines, and it deforms circles into ellipses.

- The z_1-axis is mapped to the line through $\mathbf{u}$ and the z_2-axis to the line through $\mathbf{w}$. Neither line is invariant (the flow rotates), so, unlike the real-eigenvalue case, they are a coordinate frame and not trajectories.
- A circle $|\mathbf{z}| = r_0$ becomes an ellipse through $\pm r_0\mathbf{u}$ and $\pm r_0\mathbf{w}$, with $\mathbf{u}$, $\mathbf{w}$ as *conjugate* semi-diameters. For $\rho = 0$ these ellipses are the orbits; for $\rho \neq 0$ the spirals of the $\mathbf{z}$-plane become *elliptical* spirals, tilted along $\mathbf{u}$ and $\mathbf{w}$.
- Straight lines and ratios are preserved, so the qualitative picture (spiral in, spiral out, closed loops) does not change: this is why the phase portrait can be drawn from the eigenvalues alone [Sec. 29.3, p. 291].

7.9 The sense of rotation in the $\mathbf{x}$ -plane [Sec. 29.3, Example 29.3, pp. 291–293]

In the $\mathbf{z}$-plane the rotation is clockwise by construction. A linear map with $\det Q > 0$ preserves orientation, so the rotation in the $\mathbf{x}$-plane is also clockwise; if $\det Q < 0$ the map reflects the plane and the rotation becomes **anticlockwise**. Rather than computing $\det Q$, use the vector field itself:

INFALLIBLE TEST FOR THE DIRECTION OF ROTATION

Evaluate $\dot{\mathbf{x}} = A\mathbf{x}$ at $(0,1)^T$: $A(0,1)^T = (a_{12}, a_{22})^T$. At the top of the plane, motion to the right ($a_{12} > 0$) is **clockwise**; motion to the left ($a_{12} < 0$) is **anticlockwise**. Cross-check at $(1,0)^T$: $A(1,0)^T = (a_{11}, a_{21})^T$; on the right-hand side, $a_{21} < 0$ (down) is clockwise.

The test never gives $a_{12} = 0$ for a spiral or centre: the discriminant is $\Delta = (a_{11} - a_{22})^2 + 4a_{12}a_{21} < 0$, which forces $a_{12}a_{21} < 0$, so a_{12} and a_{21} are non-zero and have opposite signs, and the two checks always agree.

7.10 Two examples solved from start to finish

EXAMPLE 7.1 (CONTINUED) – $\dot{x} = 2x + 5y$, $\dot{y} = -2x$: UNSTABLE SPIRAL

From 7.1: $\lambda = 1 \pm 3i$ ($\rho = 1, \omega = 3$), $\mathbf{u} = (5,-1)^T$, $\mathbf{w} = (0,3)^T$.

1. **Check (7.4).** $A\mathbf{u} = \begin{pmatrix} 2\cdot 5 + 5\cdot(-1) \\ -2\cdot 5 \end{pmatrix} = \begin{pmatrix} 5 \\ -10 \end{pmatrix}$ and $\rho\mathbf{u} - \omega\mathbf{w} = \begin{pmatrix} 5 \\ -1 \end{pmatrix} - 3\begin{pmatrix} 0 \\ 3 \end{pmatrix} = \begin{pmatrix} 5 \\ -10 \end{pmatrix}$. $A\mathbf{w} = \begin{pmatrix} 15 \\ 0 \end{pmatrix}$ and $\omega\mathbf{u} + \rho\mathbf{w} = 3\begin{pmatrix} 5 \\ -1 \end{pmatrix} + \begin{pmatrix} 0 \\ 3 \end{pmatrix} = \begin{pmatrix} 15 \\ 0 \end{pmatrix}$.
2. **The matrices.** $Q = \begin{pmatrix} 5 & 0 \\ -1 & 3 \end{pmatrix}$, $\det Q = 15 - 0 = 15 > 0$, and $Q^{-1} = \dfrac{1}{15}\begin{pmatrix} 3 & 0 \\ 1 & 5 \end{pmatrix}$ (swap the diagonal, change the sign of the off-diagonal, divide by the determinant). Check: $Q^{-1}Q = \dfrac{1}{15}\begin{pmatrix} 15 & 0 \\ 5-5 & 15 \end{pmatrix} = I$.

3 **Canonical form.** $AQ = [A\mathbf{u} \mid A\mathbf{w}] = \begin{pmatrix} 5 & 15 \\ -10 & 0 \end{pmatrix}$ and

$$Q^{-1}AQ = \frac{1}{15}\begin{pmatrix} 3 & 0 \\ 1 & 5 \end{pmatrix}\begin{pmatrix} 5 & 15 \\ -10 & 0 \end{pmatrix} = \frac{1}{15}\begin{pmatrix} 15 & 45 \\ 5-50 & 15 \end{pmatrix} = \begin{pmatrix} 1 & 3 \\ -3 & 1 \end{pmatrix} = J. \checkmark$$

4 **Solution in z** (7.7): $z_1 = e^t(z_{10}\cos 3t + z_{20}\sin 3t)$, $z_2 = e^t(z_{20}\cos 3t - z_{10}\sin 3t)$. Radius $r_0 e^t$, angle $\theta_0 - 3t$ (one turn every $2\pi/3 \approx 2.09$, radius multiplied by $e^{2\pi/3} \approx 8.1$ per turn).

5 **Back to x.** $x_1 = 5z_1$ and $x_2 = -z_1 + 3z_2$, so

$$x_1 = 5e^t\big(z_{10}\cos 3t + z_{20}\sin 3t\big), \qquad x_2 = e^t\big[(3z_{20} - z_{10})\cos 3t - (3z_{10} + z_{20})\sin 3t\big].$$

With $z_{10} = A$ and $z_{20} = B$ this is Robinson's formula for Example 29.1. The initial point is $\mathbf{x}_0 = (5z_{10},\ -z_{10} + 3z_{20})^T$, i.e. $(z_{10}, z_{20}) = \big(x_{10}/5,\ (x_{10} + 5x_{20})/15\big)$.

6 **Two explicit trajectories.** $\mathbf{z}_0 = (1,0)^T$ ($\mathbf{x}_0 = (5,-1)^T$): $\mathbf{x}(t) = e^t\big(5\cos 3t,\ -\cos 3t - 3\sin 3t\big)^T$. $\mathbf{z}_0 = (0,1)^T$ ($\mathbf{x}_0 = (0,3)^T$): $\mathbf{x}(t) = e^t\big(5\sin 3t,\ 3\cos 3t - \sin 3t\big)^T$. Quarter-turn check: for the second, $\mathbf{x}(\pi/6) = e^{\pi/6}(5,-1)^T = e^{\pi/6}\mathbf{u}$. The orbit passes from $\mathbf{w}$ (at $t = 0$) to $e^{\pi/6}\mathbf{u}$ a quarter of a turn later, *clockwise* in the $\mathbf{z}$-plane: z_2-axis to z_1-axis.

7 **Sense of rotation.** $\det Q = 15 > 0$ and the test of 7.9 agrees ($a_{12} = 5 > 0$): clockwise in both planes.

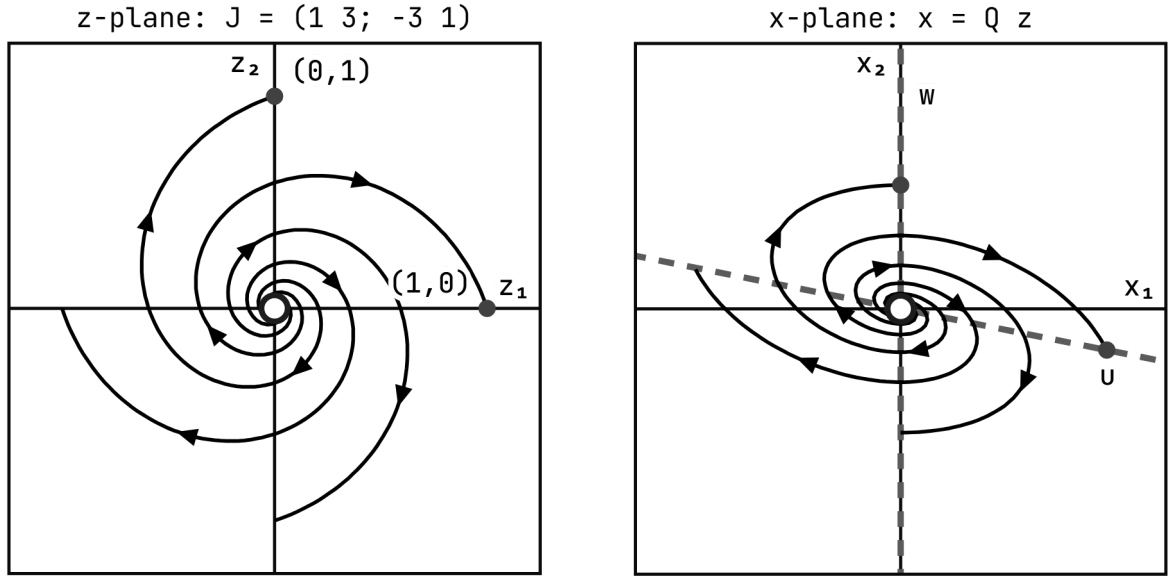


Figure 4.5. *Example 7.1, $\dot{x} = 2x + 5y$, $\dot{y} = -2x$ (unstable spiral). Left: the canonical portrait. Right: the same orbits seen in the (x_1, x_2)-plane after $\mathbf{x} = Q\mathbf{z}$; the dashed lines are the images of the z_1- and z_2-axes, i.e. the lines through $\mathbf{u}$ and $\mathbf{w}$. Both turn clockwise because* $\det Q > 0$.

THE SAME EXAMPLE IN THE FOUR STEPS OF THE NOTES: $\dot{\mathbf{x}} = A\mathbf{x}$, $A = \begin{pmatrix} 2 & 5 \\ -2 & 0 \end{pmatrix}$ [NOTES, P. 10]

1. **Step 1, eigenvalues.** $\det(A - \lambda I) = (2-\lambda)(-\lambda) + 10 = \lambda^2 - 2\lambda + 10 = 0$, $\lambda = \dfrac{2 \pm \sqrt{-36}}{2} = 1 \pm 3i$.

2. **Step 2, eigenvectors.** $\begin{pmatrix} 1 \mp 3i & 5 \\ -2 & -1 \mp 3i \end{pmatrix} \mathbf{v} = \mathbf{0}$ reduces to one row, giving $\mathbf{v}_{\pm} = \begin{pmatrix} -5 \\ 1 \mp 3i \end{pmatrix} = -\left[\begin{pmatrix} 5 \\ -1 \end{pmatrix} \pm i \begin{pmatrix} 0 \\ 3 \end{pmatrix} \right]$, a multiple of the eigenvector of 7.1, so $\mathbf{v}_1 = (5, -1)^T$, $\mathbf{v}_2 = (0, 3)^T$.

3. **Step 3, particular solution.** $\mathbf{x}(t) = r(0)\, e^t \Big(\cos\big(\theta(0) - 3t\big) \begin{pmatrix} 5 \\ -1 \end{pmatrix} + \sin\big(\theta(0) - 3t\big) \begin{pmatrix} 0 \\ 3 \end{pmatrix} \Big)$.

4. **Step 4, phase diagram.** In $\mathbf{z} = Q^{-1}\mathbf{x}$ with $Q = (\mathbf{v}_1 \mid \mathbf{v}_2)$, $\dot{\mathbf{z}} = \begin{pmatrix} 1 & 3 \\ -3 & 1 \end{pmatrix} \mathbf{z}$, whose solution $\mathbf{z}(t) = r(0) e^t \big(\cos(\theta(0) - 3t),\ \sin(\theta(0) - 3t) \big)^T$ is an unstable spiral turning clockwise ($\rho = 1 > 0, \omega = 3 > 0$). In the (x_1, x_2) axes it is still an unstable spiral; for the direction go back to the original system: on the positive x_2-axis $\dot{x}_1 = 2x_1 + 5x_2 = 5x_2 > 0$, so the motion is to the right at the top: clockwise.

EXAMPLE 7.2 – $\dot{x} = x - 2y$, $\dot{y} = x - y$: A CENTRE WHOSE ORIENTATION FLIPS

$A = \begin{pmatrix} 1 & -2 \\ 1 & -1 \end{pmatrix}$, $\operatorname{tr} A = 0$, $\det A = -1 + 2 = 1$.

1. **Eigenvalues.** $\lambda^2 + 1 = 0$, $\lambda = \pm i$: $\rho = 0, \omega = 1$. A centre.

2. **Eigenvector for $\lambda = i$.** First row: $(1 - i)v_1 - 2v_2 = 0$. Take $v_1 = 2$: $v_2 = 1 - i$. So $\mathbf{v} = (2, 1 - i)^T = \underbrace{(2, 1)^T}_{\mathbf{u}} + i\underbrace{(0, -1)^T}_{\mathbf{w}}$. Check (7.4): $A\mathbf{u} = (2 - 2,\ 2 - 1)^T = (0, 1)^T = -\mathbf{w}$ and $A\mathbf{w} = (2, 1)^T = \mathbf{u}$ (here $\rho = 0, \omega = 1$).

3. **Matrices.** $Q = \begin{pmatrix} 2 & 0 \\ 1 & -1 \end{pmatrix}$, $\det Q = -2 < 0$, $Q^{-1} = \dfrac{1}{-2} \begin{pmatrix} -1 & 0 \\ -1 & 2 \end{pmatrix} = \begin{pmatrix} 1/2 & 0 \\ 1/2 & -1 \end{pmatrix}$. Then $AQ = [A\mathbf{u} \mid A\mathbf{w}] = \begin{pmatrix} 0 & 2 \\ 1 & 1 \end{pmatrix}$ and

$$Q^{-1}AQ = \begin{pmatrix} 1/2 & 0 \\ 1/2 & -1 \end{pmatrix} \begin{pmatrix} 0 & 2 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = J. \checkmark$$

4. **Solution.** $z_1 = z_{10} \cos t + z_{20} \sin t$, $z_2 = z_{20} \cos t - z_{10} \sin t$ (circles of radius r_0, period 2π); $x_1 = 2z_1$, $x_2 = z_1 - z_2$.

5. **The orbits as curves in x.** Invert Q: $z_1 = x_1/2$, $z_2 = z_1 - x_2 = x_1/2 - x_2$. The circle $z_1^2 + z_2^2 = r_0^2$ is the ellipse

$$\tfrac{1}{4}x_1^2 + \big(\tfrac{1}{2}x_1 - x_2\big)^2 = r_0^2 \iff \tfrac{1}{2}x_1^2 - x_1x_2 + x_2^2 = r_0^2.$$

Check that $V = \tfrac{1}{2}x_1^2 - x_1x_2 + x_2^2$ is conserved: $\dot V = (x_1 - x_2)\dot x_1 + (2x_2 - x_1)\dot x_2$, and with $\dot x_1 = x_1 - 2x_2, \dot x_2 = x_1 - x_2$: $(x_1 - x_2)(x_1 - 2x_2) + (2x_2 - x_1)(x_1 - x_2) = (x_1 - x_2)\big[(x_1 - 2x_2) - (x_1 - 2x_2)\big] = 0$.

6 **Sense of rotation.** In the $\mathbf{z}$-plane: clockwise. But $\det Q = -2 < 0$. Test: $A(0,1)^T = (a_{12}, a_{22})^T = (-2,-1)^T$ points left at the top, so the motion in the $\mathbf{x}$-plane is **anticlockwise**. The reflection inside Q has reversed the orientation, which is why one must test the vector field and not copy the sign of ω.

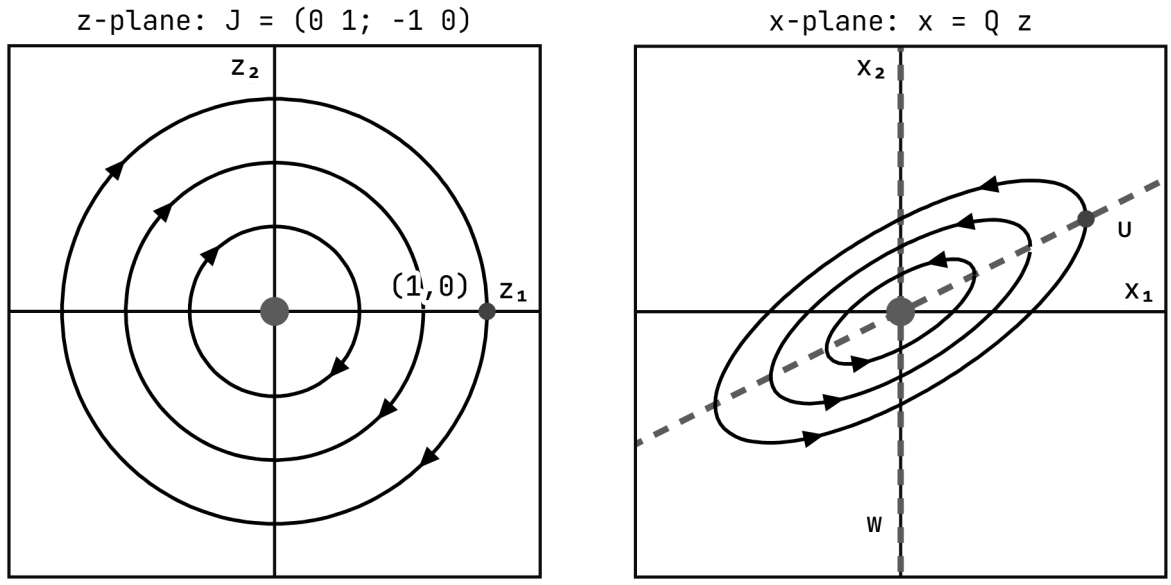


Figure 4.6. *Example 7.2, $\dot x = x - 2y$, $\dot y = x - y$ (centre). The circles of the $\mathbf{z}$-plane turn clockwise, but $\det Q = -2 < 0$ reflects the plane, so the ellipses of the $\mathbf{x}$-plane turn counter-clockwise.*

7.11 Practice: a stable spiral

EXAMPLE 7.3 — $\dot x = -5y,\ \dot y = x - 2y$ (PRACTICE: A STABLE SPIRAL)

Practice item written for this page to consolidate the complex case (it is not in the class notes). Solve the system and sketch the phase portrait in both planes.

1 **Eigenvalues.** $A = \begin{pmatrix} 0 & -5 \\ 1 & -2 \end{pmatrix}$, $\operatorname{tr} A = -2$, $\det A = 5$: $\lambda^2 + 2\lambda + 5 = 0$, $\lambda = -1 \pm 2i$ ($\rho = -1$, $\omega = 2$). A stable spiral.

2 **Eigenvector for $\lambda = -1 + 2i$.** Second row of $(A - \lambda I)\mathbf{v} = \mathbf{0}$: $v_1 + (-2 - \lambda)v_2 = v_1 - (1 + 2i)v_2 = 0$. Take $v_2 = 1$: $v_1 = 1 + 2i$. So $\mathbf{v} = (1 + 2i, 1)^T = (1,1)^T + i(2,0)^T$: $\mathbf{u} = (1,1)^T$, $\mathbf{w} = (2,0)^T$. First row check: $(0 - \lambda)(1 + 2i) - 5 = (1 - 2i)(1 + 2i) - 5 = 5 - 5 = 0$.

3 **Matrices.** $Q=\begin{pmatrix}1&2\\1&0\end{pmatrix}$, $\det Q=-2$, $Q^{-1}=\dfrac{1}{-2}\begin{pmatrix}0&-2\\-1&1\end{pmatrix}=\begin{pmatrix}0&1\\1/2&-1/2\end{pmatrix}$. $AQ=[A\mathbf{u}\mid A\mathbf{w}]=\begin{pmatrix}-5&0\\-1&2\end{pmatrix}$ and

$$Q^{-1}AQ=\begin{pmatrix}0&1\\1/2&-1/2\end{pmatrix}\begin{pmatrix}-5&0\\-1&2\end{pmatrix}=\begin{pmatrix}-1&2\\-5/2+1/2&-1\end{pmatrix}=\begin{pmatrix}-1&2\\-2&-1\end{pmatrix}.\checkmark$$

4 **Solution.** $\mathbf{z}(t)=e^{-t}\big(z_{10}\cos 2t+z_{20}\sin 2t,\ z_{20}\cos 2t-z_{10}\sin 2t\big)^T$, and $\mathbf{x}=Q\mathbf{z}=(z_1+2z_2,\ z_1)^T$. For $\mathbf{x}_0=(1,1)^T$: $\mathbf{z}_0=Q^{-1}\mathbf{x}_0=(1,0)^T$ and

$$\mathbf{x}(t)=e^{-t}\big(\cos 2t-2\sin 2t,\ \cos 2t\big)^T.$$

Check: $\dot x_1=-5x_2$ requires $\dot z_1+2\dot z_2=(-z_1+2z_2)+2(-2z_1-z_2)=-5z_1$, and $\dot x_2=\dot z_1=-z_1+2z_2=x_1-2x_2$.

5 **Sense of rotation.** Clockwise in $\mathbf{z}$, but $\det Q=-2<0$. Test: $a_{12}=-5<0$, so at $(0,1)^T$ the velocity is $(-5,-2)^T$, leftwards: **anticlockwise** in the $\mathbf{x}$-plane.

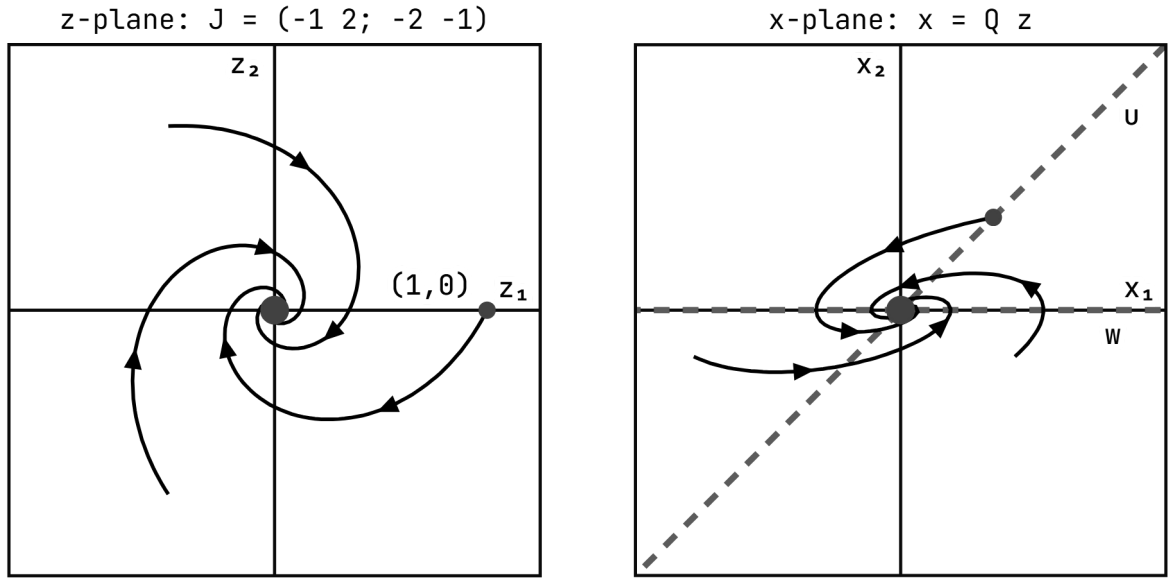


Figure 4.7. *Example 7.3, $\dot x=-5y$, $\dot y=x-2y$ (stable spiral). The spiral turns clockwise in the $\mathbf{z}$-plane and counterclockwise in the $\mathbf{x}$-plane, because $\det Q=-2<0$.*

8. A repeated eigenvalue: stars and improper nodes

CASE 3: REPEATED REAL EIGENVALUE [NOTES, P. 10]

Suppose A has only one eigenvalue λ. Two subcases: **3.1**, two different (independent) eigenvectors $\{\mathbf{v}_1, \mathbf{v}_2\}$; **3.2**, only one eigenvector $\{\mathbf{v}\}$, so A cannot be diagonalised.

8.1 Complement: algebraic and geometric multiplicity

[Sec. 30, p. 295 and Example 27.4, p. 265]

The characteristic polynomial of a 2×2 matrix has a double root exactly when $\Delta = (\operatorname{tr} A)^2 - 4 \det A = 0$. Then

$$\lambda_1 = \lambda_2 = \lambda = \tfrac{1}{2} \operatorname{tr} A, \qquad \det(A - \mu I) = (\mu - \lambda)^2.$$

- The **algebraic multiplicity** of λ is the multiplicity of the root: $m_a = 2$.
- The **geometric multiplicity** is the number of independent eigenvectors: $m_g = \dim \ker(A - \lambda I)$. It is at least 1 (an eigenvalue has an eigenvector) and at most m_a, so $m_g \in \{1, 2\}$.

The two values of m_g produce two very different phase portraits, and no other possibility exists for a real 2×2 matrix with $\Delta = 0$.

8.2 Subcase 3.1, $m_g = 2$: the stars [Sec. 30.1, p. 295]

SUBCASE 3.1: TWO DIFFERENT EIGENVECTORS $\{\mathbf{v}_1, \mathbf{v}_2\}$ [NOTES, P. 10]

We can diagonalise A as $A = PDP^{-1}$, where $P = (\mathbf{v}_1, \mathbf{v}_2)$ and $D = \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix} = \lambda I$. Then $A = P(\lambda I)P^{-1} = \lambda P P^{-1} = \lambda I$, so $(A - \lambda I)\mathbf{v} = 0\,\mathbf{v} = \mathbf{0}$ for every $\mathbf{v} \in \mathbb{R}^2$: every vector is an eigenvector, and $\mathbf{x}(t) = \mathbf{x}_0 e^{\lambda t}$ is a particular solution for every $\mathbf{x}_0$. Phase diagrams: **unstable star** ($\lambda > 0$) and **stable star** ($\lambda < 0$); every line through the origin is a solution.

PROPOSITION – TWO INDEPENDENT EIGENVECTORS FORCE $A = \lambda I$

If λ is a repeated eigenvalue of the 2×2 matrix A and $m_g = 2$, then $A = \lambda I = \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix}$.

PROOF

Let $\mathbf{v}_1, \mathbf{v}_2$ be independent eigenvectors, $A\mathbf{v}_i = \lambda \mathbf{v}_i$. Put $B = [\mathbf{v}_1 \mid \mathbf{v}_2]$, invertible. Then $AB = [\lambda \mathbf{v}_1 \mid \lambda \mathbf{v}_2] = \lambda B$, and multiplying on the right by B^{-1} gives $A = \lambda B B^{-1} = \lambda I$. (Equivalently, $\ker(A - \lambda I)$ is a two-dimensional subspace of $\mathbb{R}^2$, so it is all of $\mathbb{R}^2$, and $(A - \lambda I)\mathbf{x} = \mathbf{0}$ for every $\mathbf{x}$ means $A - \lambda I = O$.)

■

So every vector is an eigenvector, and the system uncouples in *every* basis: $\dot{x}_1 = \lambda x_1, \dot{x}_2 = \lambda x_2$. For any change of basis, $Q^{-1}(\lambda I)Q = \lambda I$, so there is no separate $\mathbf{z}$-plane. Integrate each equation, $\dfrac{d}{dt}(e^{-\lambda t}x_i) = 0$, and then

$$\mathbf{x}(t) = e^{\lambda t}\mathbf{x}_0 = e^{\lambda t}(c_1\mathbf{v}_1 + c_2\mathbf{v}_2) \quad \text{for any basis } \mathbf{v}_1, \mathbf{v}_2. \tag{8.1}$$

The orbit through $\mathbf{x}_0$ stays on the straight line through the origin and $\mathbf{x}_0$: **every line through the origin is invariant**, and the point moves along it with speed proportional to its distance from the origin.

- **Stable star**, $\lambda < 0$: all solutions go to $\mathbf{0}$ along straight rays (the origin is asymptotically stable).
- **Unstable star**, $\lambda > 0$: all solutions leave $\mathbf{0}$ along straight rays.
- For $\lambda = 0$ we get $A = O$: every point is an equilibrium (Robinson Exercise 30.5).

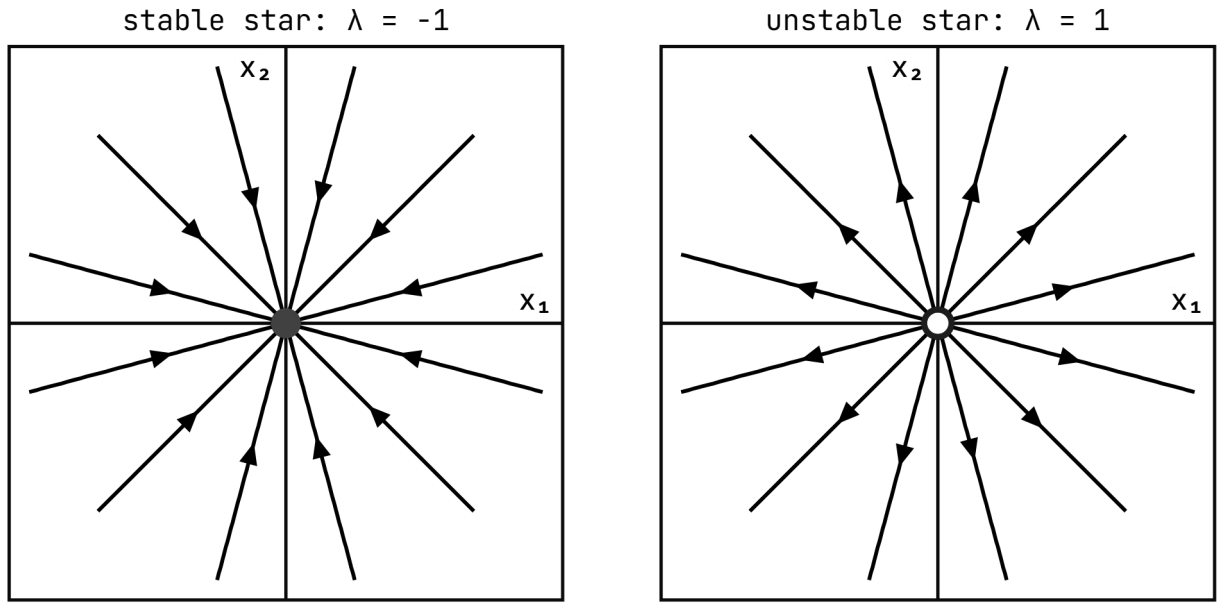


Figure 4.8. *$A = \lambda I$: every line through the origin is invariant and every solution moves along its own line. The picture is the same in any basis, so there is no separate $\mathbf{z}$-plane.*

8.3 Subcase 3.2, $m_g = 1$: the construction of the notes

[Notes, p. 10; Robinson, Sec. 30.2, pp. 295–297]

Now $A \neq \lambda I$, there is a single eigendirection $\mathbf{v}$, and (8.1) cannot give the general solution (it has only one free constant). The notes build a second basis vector as follows [Notes, p. 10].

1. **Choose and define.** Choose any $\mathbf{v}_2$ linearly independent of $\mathbf{v}$, and define $\mathbf{v}_1 = (A - \lambda I)\mathbf{v}_2$. $\mathbf{v}_1 \neq \mathbf{0}$: otherwise $\mathbf{v}_2$ would be a second, independent eigenvector.

2. **Write $\mathbf{v}_1$ in the basis $\{\mathbf{v}_2, \mathbf{v}\}$ of $\mathbb{R}^2$:** $\mathbf{v}_1 = \alpha\mathbf{v}_2 + \beta\mathbf{v}$.

3 **Apply $A-\lambda I$.** Since $(A-\lambda I)\mathbf{v}=\mathbf{0}$,

$$(A-\lambda I)\mathbf{v}_1=(A-\lambda I)(\alpha\mathbf{v}_2+\beta\mathbf{v})=\alpha(A-\lambda I)\mathbf{v}_2=\alpha\mathbf{v}_1 \quad\Longrightarrow\quad \big(A-(\alpha+\lambda)I\big)\mathbf{v}_1=\mathbf{0}.$$

4 **Conclude $\alpha=0$.** $\mathbf{v}_1\neq\mathbf{0}$ is then an eigenvector for $\lambda+\alpha$; but the only eigenvalue is λ, so $\alpha=0$. Hence $\mathbf{v}_1=\beta\mathbf{v}$ is proportional to $\mathbf{v}$ (an eigenvector, $(A-\lambda I)\mathbf{v}_1=\mathbf{0}$), and $\{\mathbf{v}_1,\mathbf{v}_2\}$ is a basis.

5 **The relation between them.** By definition $(A-\lambda I)\mathbf{v}_2=\mathbf{v}_1$, i.e.

$$A\mathbf{v}_2=\mathbf{v}_1+\lambda\mathbf{v}_2. \tag{8.2}$$

6 **The matrix.** Let $P=(\mathbf{v}_1,\mathbf{v}_2)$. Then $AP=(A\mathbf{v}_1,A\mathbf{v}_2)=(\lambda\mathbf{v}_1,\ \mathbf{v}_1+\lambda\mathbf{v}_2)=(\mathbf{v}_1,\mathbf{v}_2)\begin{pmatrix}\lambda & 1\\ 0 & \lambda\end{pmatrix}$, so

$$A=P\begin{pmatrix}\lambda & 1\\ 0 & \lambda\end{pmatrix}P^{-1}.$$

$\mathbf{v}_1$ is the eigenvector and $\mathbf{v}_2$ a **generalised eigenvector**. Robinson builds the pair in the same order (p. 296). An equivalent route, often quicker by hand, starts from the eigenvector and solves for $\mathbf{v}_2$; it rests on the following lemma (complement).

LEMMA – $N^2=O$ [CAYLEY-HAMILTON; EXERCISES 30.2-30.3]
If $\det(A-\mu I)=(\mu-\lambda)^2$, then $N=A-\lambda I$ satisfies $N^2=O$.

PROOF
The Cayley–Hamilton theorem for 2×2 matrices says that A satisfies its own characteristic equation $\mu^2-(\operatorname{tr}A)\mu+\det A=0$, i.e. $A^2-(\operatorname{tr}A)A+(\det A)I=O$. Here $\operatorname{tr}A=2\lambda$ and $\det A=\lambda^2$, so $A^2-2\lambda A+\lambda^2 I=(A-\lambda I)^2=N^2=O$.

■

The eigenvector-first route produces the same two vectors, linked as a *chain*:

1 An eigenvector $\mathbf{v}_1\neq\mathbf{0}$: $N\mathbf{v}_1=\mathbf{0}$, i.e. $A\mathbf{v}_1=\lambda\mathbf{v}_1$.

2 A **generalised eigenvector** $\mathbf{v}_2$ with $N\mathbf{v}_2=\mathbf{v}_1$, i.e. $(A-\lambda I)\mathbf{v}_2=\mathbf{v}_1$, equivalently

$$A\mathbf{v}_2=\lambda\mathbf{v}_2+\mathbf{v}_1. \tag{8.2}$$

Why $\mathbf{v}_2$ exists. Since $N^2 = O$, every vector $N\mathbf{y}$ satisfies $N(N\mathbf{y}) = \mathbf{0}$, so the image of N lies inside its kernel. Both are one-dimensional ($\dim\ker N = m_g = 1$ and $\dim\operatorname{im} N = 2 - 1 = 1$), so they coincide: $\operatorname{im} N = \ker N$. Hence the eigenvector $\mathbf{v}_1 \in \ker N$ is in the image of N and the system $N\mathbf{v}_2 = \mathbf{v}_1$ is solvable (its two equations are proportional: N has rank 1). The solution is not unique: $\mathbf{v}_2 + s\mathbf{v}_1$ works for every s, because $N\mathbf{v}_1 = \mathbf{0}$. Rescaling $\mathbf{v}_1$ rescales $\mathbf{v}_2$ in the same way.

Why $\mathbf{v}_1, \mathbf{v}_2$ are a basis. If $a\mathbf{v}_1 + b\mathbf{v}_2 = \mathbf{0}$, apply N: $bN\mathbf{v}_2 = b\mathbf{v}_1 = \mathbf{0}$, so $b = 0$; then $a\mathbf{v}_1 = \mathbf{0}$ gives $a = 0$.

> TWO ROUTES, ONE CHAIN
>
> The notes and Robinson (p. 296) pick *any* $\mathbf{v}_2$ not parallel to the eigenvector and set $\mathbf{v}_1 = (A - \lambda I)\mathbf{v}_2$; with the lemma, $N\mathbf{v}_1 = N^2\mathbf{v}_2 = \mathbf{0}$ shows at once that $\mathbf{v}_1$ is an eigenvector (the notes prove it with the $\alpha = 0$ argument instead). The eigenvector-first order, used in Example 8.1 and in the algorithm at the end, gives the same chain up to the choice of $\mathbf{v}_2$.

8.4 The change of variable $\mathbf{z} = Q^{-1}\mathbf{x}$ and the Jordan block

[Sec. 30.2, pp. 296-297, Eqs. (30.2)-(30.3)]

Take $Q = [\,\mathbf{v}_1 \mid \mathbf{v}_2\,]$ (eigenvector first). Column by column, with (8.2):

$$AQ = [A\mathbf{v}_1 \mid A\mathbf{v}_2] = [\lambda\mathbf{v}_1 \mid \mathbf{v}_1 + \lambda\mathbf{v}_2] = [\mathbf{v}_1 \mid \mathbf{v}_2]\begin{pmatrix}\lambda & 1\\ 0 & \lambda\end{pmatrix} = Q\,J.$$

(First column of J: $(\lambda, 0)^T$ gives $\lambda\mathbf{v}_1$. Second column: $(1, \lambda)^T$ gives $1\cdot\mathbf{v}_1 + \lambda\mathbf{v}_2$.) Multiplying by Q^{-1} on the left:

$$\boxed{Q^{-1}AQ = J = \begin{pmatrix}\lambda & 1\\ 0 & \lambda\end{pmatrix}, \qquad \dot z_1 = \lambda z_1 + z_2, \quad \dot z_2 = \lambda z_2.} \tag{8.3}$$

This is **partially** uncoupled: the second equation involves only z_2, the first is driven by it. The notes call the matrix P and define $\mathbf{z} = P^{-1}\mathbf{x}$, so that $\dot{\mathbf{z}} = P^{-1}A\mathbf{x} = P^{-1}P\begin{pmatrix}\lambda & 1\\ 0 & \lambda\end{pmatrix}P^{-1}\mathbf{x} = \begin{pmatrix}\lambda & 1\\ 0 & \lambda\end{pmatrix}\mathbf{z}$ [Notes, p. 10].

8.5 Solving for $\mathbf{z}$ [Sec. 30.2, p. 297, Eq. (30.4)]

1. **The second equation.** $\dot z_2 = \lambda z_2$. Multiply by $e^{-\lambda t}$: $\dfrac{d}{dt}\left(e^{-\lambda t}z_2\right) = e^{-\lambda t}(\dot z_2 - \lambda z_2) = 0$, so $e^{-\lambda t}z_2$ is constant: $z_2(t) = c_2 e^{\lambda t}$ with $c_2 = z_2(0)$.

2. **The first equation becomes a forced linear equation.** Substituting z_2: $\dot z_1 - \lambda z_1 = c_2 e^{\lambda t}$. The integrating factor is $e^{-\lambda t}$ (its derivative is $-\lambda e^{-\lambda t}$, which is exactly the coefficient needed).

3 **Multiply and recognise a derivative.** $e^{-\lambda t}\dot z_1 - \lambda e^{-\lambda t} z_1 = \dfrac{d}{dt}\big(e^{-\lambda t} z_1\big)$ by the product rule, and the right side is $e^{-\lambda t} c_2 e^{\lambda t} = c_2$. So

$$\frac{d}{dt}\big(e^{-\lambda t} z_1\big) = c_2 .$$

4 **Integrate with Barrow's rule** between 0 and t: $\big[e^{-\lambda s} z_1(s)\big]_{s=0}^{s=t} = \displaystyle\int_0^t c_2\, ds = c_2 t$, i.e. $e^{-\lambda t} z_1(t) - z_1(0) = c_2 t$. With $c_1 = z_1(0)$:

$$z_1(t) = (c_1 + c_2 t) e^{\lambda t}, \qquad z_2(t) = c_2 e^{\lambda t}. \tag{8.4}$$

5 **Check.** $\dot z_1 = c_2 e^{\lambda t} + \lambda (c_1 + c_2 t) e^{\lambda t} = \lambda z_1 + z_2$.

In matrix form $\mathbf{z}(t) = e^{\lambda t} \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix} \mathbf{z}_0$ (this is e^{Jt}, since $J = \lambda I + \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ and the second matrix squares to zero). The factor t in z_1 is the vector version of the $te^{\lambda t}$ of the repeated characteristic root in Topic 3.

8.6 The general solution in $\mathbf{x}$ [Sec. 30.2, p. 297, Eq. (30.4)]

Undo the change of basis, $\mathbf{x} = Q\mathbf{z} = z_1 \mathbf{v}_1 + z_2 \mathbf{v}_2$:

$$\mathbf{x}(t) = (c_1 + c_2 t) e^{\lambda t} \mathbf{v}_1 + c_2 e^{\lambda t} \mathbf{v}_2 \;=\; \boxed{\,e^{\lambda t}\Big[c_1 \mathbf{v}_1 + c_2\big(t\,\mathbf{v}_1 + \mathbf{v}_2\big)\Big]\,} \tag{8.5}$$

- $c_1 e^{\lambda t}\mathbf{v}_1$: the pure eigenmode, a motion along the line span $\mathbf{v}_1$.
- $c_2 e^{\lambda t}\mathbf{v}_2$: the generalised-eigenvector mode; on its own it is not a solution.
- $c_2 t e^{\lambda t}\mathbf{v}_1$: the *secular term*. It couples the two modes and is the only reason the orbit is not a straight line: as $t \to \pm\infty$ it dominates, so the orbits become parallel to $\mathbf{v}_1$ at both ends.
- The constants are the initial coordinates in the chain basis: $(c_1, c_2)^T = Q^{-1}\mathbf{x}_0$.

The notes write the same result with $C = c_2$ and $D = c_1$: $\mathbf{x} = P\mathbf{z} = (\mathbf{v}_1, \mathbf{v}_2)\,\mathbf{z} = (Ct + D)\, e^{\lambda t}\mathbf{v}_1 + C\, e^{\lambda t}\mathbf{v}_2$ [Notes, p. 10].

Direct check. With $\mathbf{x} = e^{\lambda t}\mathbf{s}(t)$, $\mathbf{s} = c_1\mathbf{v}_1 + c_2(t\mathbf{v}_1 + \mathbf{v}_2)$: $\dot{\mathbf{x}} = \lambda\mathbf{x} + e^{\lambda t} c_2 \mathbf{v}_1$. And by (8.2), $A\mathbf{s} = c_1\lambda\mathbf{v}_1 + c_2\big(t\lambda\mathbf{v}_1 + \lambda\mathbf{v}_2 + \mathbf{v}_1\big) = \lambda\mathbf{s} + c_2\mathbf{v}_1$, so $A\mathbf{x} = \lambda\mathbf{x} + e^{\lambda t}c_2\mathbf{v}_1 = \dot{\mathbf{x}}$. The two solutions $e^{\lambda t}\mathbf{v}_1$ and $e^{\lambda t}(t\mathbf{v}_1 + \mathbf{v}_2)$ have Wronskian $e^{2\lambda t}\det[\mathbf{v}_1\ \mathbf{v}_2] \neq 0$, so they form a fundamental set.

8.7 The improper node in the $\mathbf{z}$ -plane [Sec. 30.2, pp. 297–298]

Eliminate t from (8.4) for $c_2 \neq 0$: since $z_2 = c_2 e^{\lambda t}$, $t = \frac{1}{\lambda}\ln(z_2/c_2)$ and $c_2 e^{\lambda t} = z_2$, so

$$z_1 = \Big(\frac{c_1}{c_2} + t\Big) z_2 = z_2\Big[K + \frac{1}{\lambda}\ln|z_2|\Big], \qquad K = \frac{c_1}{c_2} - \frac{1}{\lambda}\ln|c_2|. \tag{8.6}$$

(The same curve from the slope: $\dfrac{dz_1}{dz_2} = \dfrac{\dot z_1}{\dot z_2} = \dfrac{\lambda z_1 + z_2}{\lambda z_2} = \dfrac{z_1}{z_2} + \dfrac{1}{\lambda}$, so $\dfrac{d}{dz_2}\dfrac{z_1}{z_2} = \dfrac{1}{\lambda z_2}$ and $\dfrac{z_1}{z_2} = \dfrac{1}{\lambda}\ln|z_2| + K$.)

- **The z_1-axis is invariant.** For $c_2 = 0$: $z_2 \equiv 0$ and $z_1 = c_1 e^{\lambda t}$, two half-lines moving away from the origin if $\lambda > 0$ (towards it if $\lambda < 0$). This is the single eigendirection, span $\mathbf{v}_1$.
- **Where the orbits turn.** The notes start the sketch from the line where $\dot z_1 = 0$: $\lambda z_1 + z_2 = 0$, i.e. $z_2 = -\lambda z_1$ [Notes, p. 10]. There the velocity $(0, \lambda z_2)$ is vertical, so every S-shaped orbit crosses this line with a vertical tangent, where z_1 reaches its extreme value.
- **Direction of motion.** $\dot z_1 = \lambda z_1 + z_2$ equals z_2 on the z_2-axis. So in the upper half-plane ($z_2 > 0$) orbits cross the z_2-axis moving *right* and in the lower half-plane moving *left*.
- **Behaviour as $t \to \pm\infty$** (take $\lambda > 0$): for $t \to -\infty$ the orbit tends to the origin and $z_1/z_2 = c_1/c_2 + t \to \mp\infty$, so it arrives *tangent to the z_1-axis*. For $t \to +\infty$ the orbit escapes to infinity with $z_1/z_2 \sim t \to \pm\infty$ again: it becomes parallel to the z_1-axis. For $\lambda < 0$ everything is reversed in time.
- **Shape.** Near the origin $z_1 \approx (z_2/\lambda)\ln z_2$, which tends to 0 with infinite slope $dz_1/dz_2 \to -\infty$ (for $\lambda > 0$): the orbit leaves the origin horizontally, curls over the z_2-axis and flattens again. Upper and lower half-planes are point-symmetric ($\mathbf{z} \mapsto -\mathbf{z}$ maps solutions to solutions), giving an S-shaped pair of curves. The approach to the axis is only logarithmic ($z_1/z_2 = K + \frac{1}{\lambda}\ln|z_2|$ grows very slowly as $z_2 \to 0$), so at the scale of Fig. 4.9 the portrait looks like a fan of almost straight rays that bend sharply only close to the origin.

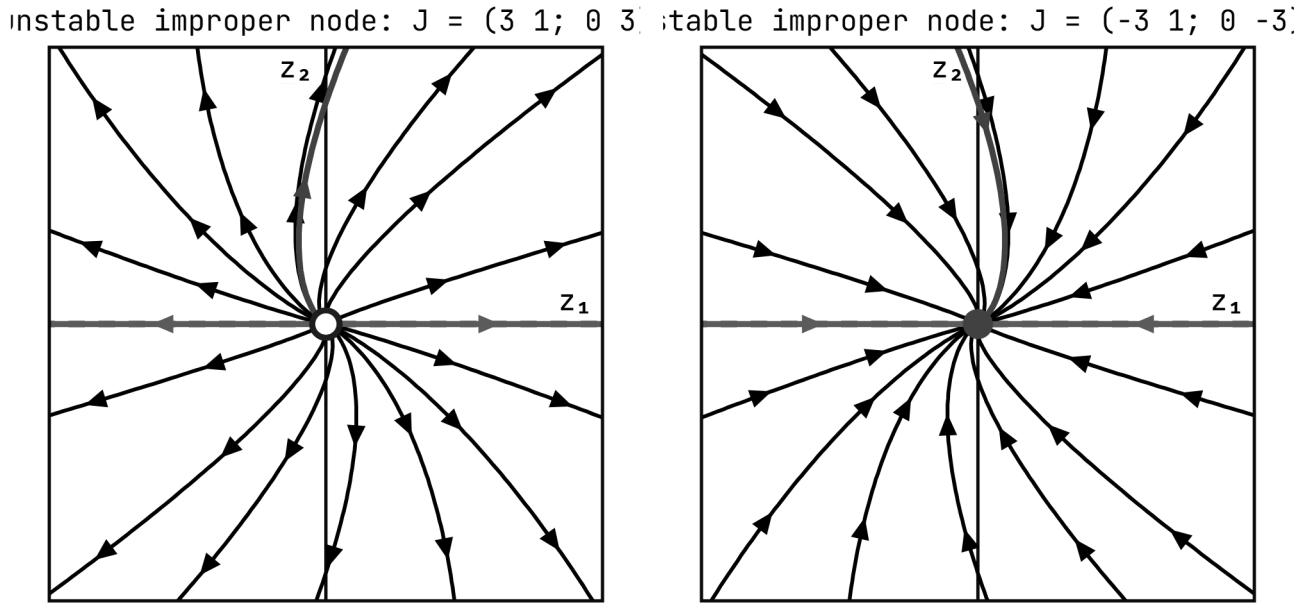


Figure 4.9. *The canonical improper node in the $\mathbf{z}$-plane. Amber: the invariant z_1-axis (the only eigendirection). Green: the orbit through $(0, 1)$. All orbits are tangent to the z_1-axis at the origin and become parallel to it at infinity; above the axis they move right, below it left.*

8.8 The improper node in the $\mathbf{x}$ -plane [Sec. 30.2, p. 298]

The map $\mathbf{x} = Q\mathbf{z}$ sends the z_1-axis to the line span $\mathbf{v}_1$, the only invariant line, and the z_2-axis to span $\mathbf{v}_2$ (a reference direction, not an orbit). Straight lines go to straight lines, so the properties of the $\mathbf{z}$-portrait that are stated with lines survive:

- All orbits are tangent to span $\mathbf{v}_1$ at the origin, and become parallel to it far away, bending in an S. In the words of the notes: the same as in $\mathbf{z}$, but with the trajectories drawn towards $\mathbf{v}_1$; the exact shape needs an explicit A [Notes, p. 10].
- The role of $\mathbf{v}_2$ is to say *which side* and *which way*. Write $\mathbf{x} = z_1\mathbf{v}_1 + z_2\mathbf{v}_2$: the sign of z_2 tells on which side of the line span $\mathbf{v}_1$ the point lies (the side where $\mathbf{v}_2$ points if $z_2 > 0$). The motion on that side is along $+\mathbf{v}_1$ for $z_2 > 0$, along $-\mathbf{v}_1$ for $z_2 < 0$ (because $\dot{z}_1 = \lambda z_1 + z_2$).
- Practical rule, independent of the algebra: $\dot{\mathbf{x}} - \lambda\mathbf{x} = (A - \lambda I)\mathbf{x} = N\mathbf{x} = z_2\mathbf{v}_1$. At any point off the eigenline, compute $N\mathbf{x}$: it is a multiple of $\mathbf{v}_1$ whose sign says how the orbit drifts along the eigendirection there.
- Because Q may reflect ($\det Q < 0$) the S can look mirrored with respect to the $\mathbf{z}$-plane; the rule above is what fixes the orientation in the $\mathbf{x}$-plane.

8.9 A complete worked example [Example 30.1, p. 298]

EXAMPLE 8.1 — $\dot{x} = 5x - 4y$, $\dot{y} = x + y$: UNSTABLE IMPROPER NODE

$A = \begin{pmatrix} 5 & -4 \\ 1 & 1 \end{pmatrix}$.

1. **Eigenvalue.** $\operatorname{tr} A = 6$, $\det A = 5 + 4 = 9$, so $\lambda^2 - 6\lambda + 9 = (\lambda - 3)^2 = 0$: $\lambda = 3$ with $m_a = 2$ ($\Delta = 36 - 36 = 0$).

2. **Geometric multiplicity.** $N = A - 3I = \begin{pmatrix} 2 & -4 \\ 1 & -2 \end{pmatrix} \neq O$ has rank 1 (the rows are proportional), so $m_g = 1$: an improper node, not a star.

3. **Eigenvector.** $N\mathbf{v} = \mathbf{0}$: $2a - 4b = 0$, $a = 2b$. Take $\mathbf{v}_1 = (2, 1)^T$.

4. **Generalised eigenvector.** $N\mathbf{v}_2 = \mathbf{v}_1$: $2a - 4b = 2$ and $a - 2b = 1$ (the same equation). Take $b = 0, a = 1$: $\mathbf{v}_2 = (1, 0)^T$. (General solution $\mathbf{v}_2 = (1, 0)^T + s(2, 1)^T$.)

5. **Matrices.** $Q = \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix}$, $\det Q = 0 - 1 = -1$, $Q^{-1} = \dfrac{1}{-1}\begin{pmatrix} 0 & -1 \\ -1 & 2 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & -2 \end{pmatrix}$. Check: $Q^{-1}Q = \begin{pmatrix} 0+1 & 0 \\ 2-2 & 1 \end{pmatrix} = I$.

6. **Jordan form.** $A\mathbf{v}_1 = (10 - 4,\ 2 + 1)^T = (6, 3)^T = 3\mathbf{v}_1$, $A\mathbf{v}_2 = (5, 1)^T = \mathbf{v}_1 + 3\mathbf{v}_2 = (2 + 3,\ 1 + 0)^T$. So $AQ = \begin{pmatrix} 6 & 5 \\ 3 & 1 \end{pmatrix}$ and

$$Q^{-1}AQ = \begin{pmatrix} 0 & 1 \\ 1 & -2 \end{pmatrix}\begin{pmatrix} 6 & 5 \\ 3 & 1 \end{pmatrix} = \begin{pmatrix} 3 & 1 \\ 6-6 & 5-2 \end{pmatrix} = \begin{pmatrix} 3 & 1 \\ 0 & 3 \end{pmatrix} = J. \checkmark$$

7. **Solution in $\mathbf{z}$** (8.4): $z_1 = (c_1 + c_2 t)e^{3t}$, $z_2 = c_2 e^{3t}$.

8 **Solution in x** (8.5): $\mathbf{x}(t) = e^{3t}\Big[c_1\begin{pmatrix}2\\1\end{pmatrix} + c_2\Big(t\begin{pmatrix}2\\1\end{pmatrix} + \begin{pmatrix}1\\0\end{pmatrix}\Big)\Big]$, i.e.

$$x_1 = e^{3t}\big[2c_1 + c_2(2t+1)\big], \qquad x_2 = e^{3t}\big[c_1 + c_2 t\big].$$

Initial data: $\mathbf{x}_0 = (2c_1 + c_2,\ c_1)^T$, so $c_1 = x_{20}$ and $c_2 = x_{10} - 2x_{20}$.

9 **Direct check.** $\dot{x}_1 = e^{3t}\{3[2c_1 + c_2(2t+1)] + 2c_2\} = e^{3t}[6c_1 + c_2(6t+5)]$, and $5x_1 - 4x_2 = e^{3t}[10c_1 + c_2(10t+5) - 4c_1 - 4c_2 t] = e^{3t}[6c_1 + c_2(6t+5)]$. $\dot{x}_2 = e^{3t}[3(c_1 + c_2 t) + c_2]$ and $x_1 + x_2 = e^{3t}[3c_1 + c_2(3t+1)]$.

Explicit trajectories.

(c_1, c_2)	$\mathbf{x}_0$	$\mathbf{x}(t)$	ORBIT
$(1, 0)$	$(2, 1)$	$e^{3t}(2, 1)^T$	half-line of span $\mathbf{v}_1$, outward
$(-1, 0)$	$(-2, -1)$	$-e^{3t}(2, 1)^T$	opposite half-line, outward
$(0, 1)$	$(1, 0)$	$e^{3t}(2t+1,\ t)^T$	crosses $x_1 = 0$ at $t = -\frac{1}{2}$, at $e^{-3/2}(0, -\frac{1}{2})$
$(0, -1)$	$(-1, 0)$	$-e^{3t}(2t+1,\ t)^T$	point-symmetric to the previous one
$(1, 1)$	$(3, 1)$	$e^{3t}(2t+3,\ t+1)^T$	same side of span $\mathbf{v}_1$ as $\mathbf{v}_2$

Asymptotics. For $c_2 \neq 0$, as $t \to -\infty$: $\mathbf{x}(t) \approx c_2 t e^{3t}\mathbf{v}_1 \to \mathbf{0}$ (tangent to span $\mathbf{v}_1$, from the side $-\operatorname{sign}(c_2)\mathbf{v}_1$); as $t \to +\infty$: $|\mathbf{x}| \to \infty$ with direction $\to +\operatorname{sign}(c_2)\mathbf{v}_1$. For example, $\mathbf{x}(t) = e^{3t}(2t+1, t)^T$ arrives at the origin along the direction $-\mathbf{v}_1$ and leaves along $+\mathbf{v}_1$.

Orientation. The side of the eigenline is given by $c_2 = x_1 - 2x_2$ (positive below the line $x_1 = 2x_2$, on the side where $\mathbf{v}_2$ lies) and the drift by $N\mathbf{x} = c_2\mathbf{v}_1$. At $(0, 1)^T$: $N(0, 1)^T = (-4, -2)^T = -2\mathbf{v}_1$, leftwards, in agreement with $\dot{x} = -4y < 0$ on the positive y-axis.

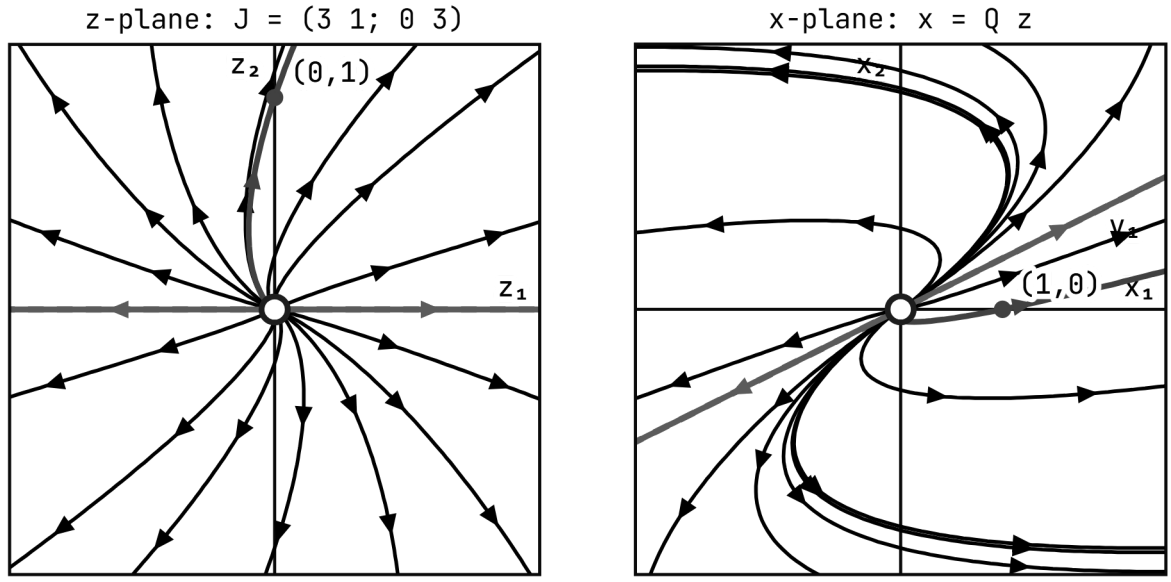


Figure 4.10. *Example 8.1, $\dot{x} = 5x - 4y$, $\dot{y} = x + y$ (unstable improper node, $\lambda = 3$). Left: canonical portrait. Right: the same orbits in the (x_1, x_2)-plane, with $\mathbf{v}_1 = (2,1)^T$ and $\mathbf{v}_2 = (1,0)^T$. The amber line is* $\operatorname{span}\mathbf{v}_1$*; the green orbit is $\mathbf{x}(t) = e^{3t}(2t+1,\ t)^T$. Because $\det Q = -1$, the map reverses orientation, so the S is traversed in the opposite sense to the one in the $\mathbf{z}$-plane.*

The stable twin. Reversing time turns an improper node into its stable twin: $A' = -A = \begin{pmatrix} -5 & 4 \\ -1 & -1 \end{pmatrix}$ has $\lambda' = -3$, the same eigenvector $\mathbf{v}_1 = (2,1)^T$, and $N' = A' + 3I = -N$, so $N'\mathbf{v}_2' = \mathbf{v}_1$ is solved by $\mathbf{v}_2' = -\mathbf{v}_2 = (-1,0)^T$. Then $Q' = [\mathbf{v}_1 \mid -\mathbf{v}_2]$, $J' = \begin{pmatrix} -3 & 1 \\ 0 & -3 \end{pmatrix}$ and the orbits are the previous curves with all arrows reversed.

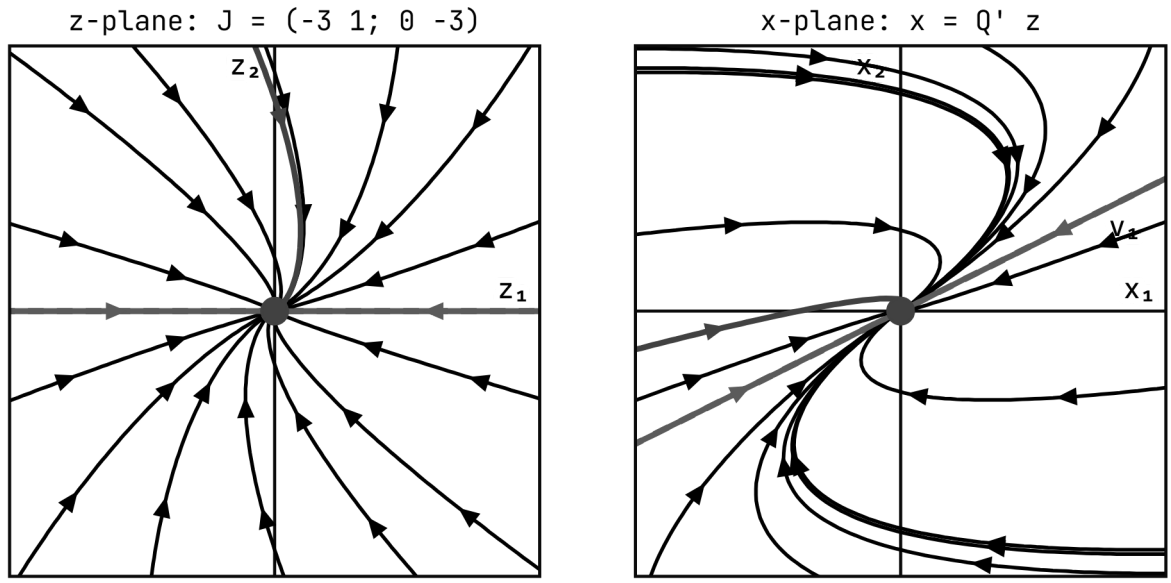


Figure 4.11. *The stable twin $A' = -A = \begin{pmatrix} -5 & 4 \\ -1 & -1 \end{pmatrix}$ ($\lambda = -3$, $Q' = [\mathbf{v}_1 \mid -\mathbf{v}_2]$): the same curves with every arrow reversed.*

ALGORITHM FOR $\dot{\mathbf{x}} = A\mathbf{x}$ WITH $\Delta = 0$

1. $\lambda = \frac{1}{2}\operatorname{tr} A$. Compute $N = A - \lambda I$.
2. If $N = O$: $A = \lambda I$, a star (stable if $\lambda < 0$, unstable if $\lambda > 0$); $\mathbf{x}(t) = e^{\lambda t}\mathbf{x}_0$.
3. Otherwise: $\mathbf{v}_1$ spans $\ker N$; solve $N\mathbf{v}_2 = \mathbf{v}_1$; $Q = [\mathbf{v}_1 \mid \mathbf{v}_2]$.
4. $\mathbf{x}(t) = e^{\lambda t}\left[c_1\mathbf{v}_1 + c_2(t\mathbf{v}_1 + \mathbf{v}_2)\right]$ with $(c_1, c_2)^T = Q^{-1}\mathbf{x}_0$; improper node, stable if $\lambda < 0$, unstable if $\lambda > 0$.
5. Draw span $\mathbf{v}_1$ with arrows (outward if $\lambda > 0$), then the S-shaped orbits tangent to it; fix the side and the orientation with $N\mathbf{x}$ at one point off the line.

SECTION 09 • ROBINSON CH. 31

9. Classification of linear phase portraits

Robinson's summary [Ch. 31, p. 301]: compute the eigenvalues of A; if they are real, compute the eigenvectors.

EIGENVALUES	CONDITION	ORIGIN IS A	BEHAVIOUR
Real, distinct	$\lambda_1 < \lambda_2 < 0$	stable node	All trajectories approach $\mathbf{0}$, tangent to the eigenvector of λ_2 (smaller $\lvert\lambda\rvert$)
	$\lambda_1 > \lambda_2 > 0$	unstable node	All trajectories leave $\mathbf{0}$, tangent to the eigenvector of λ_2
	$\lambda_1 < 0 < \lambda_2$	saddle	Only trajectories on the stable eigenvector reach $\mathbf{0}$
Complex $\rho \pm i\omega$	$\rho < 0$	stable spiral	Spiral in; sense from the sign of $\dot{x}$ on $x = 0$
	$\rho > 0$	unstable spiral	Spiral out
	$\rho = 0$	centre	Closed periodic orbits
Repeated λ	$A = \lambda I$	star	Straight lines; stable if $\lambda < 0$
	$A \neq \lambda I$	improper node	S-shaped; stable if $\lambda < 0$

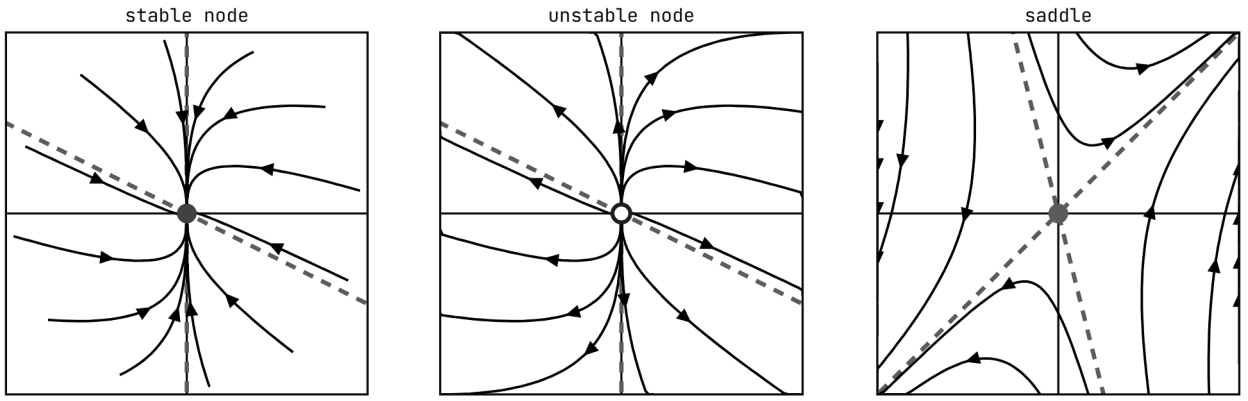


Figure 4.12 (a). *The nine linear phase portraits, arranged as in Robinson Fig. 31.1 (p. 302). First row: real distinct eigenvalues (stable node, unstable node, saddle).*

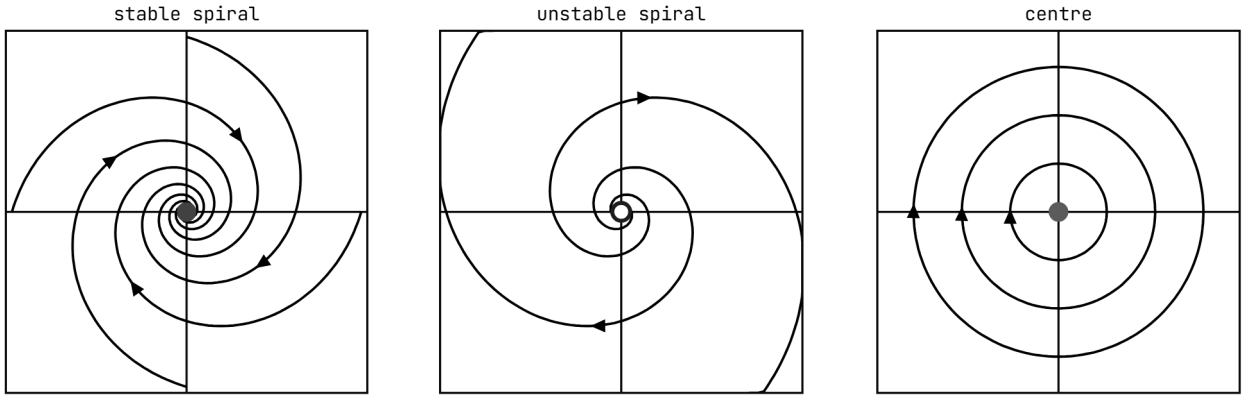


Figure 4.12 (b). *Second row: complex eigenvalues (stable spiral, unstable spiral, centre).*

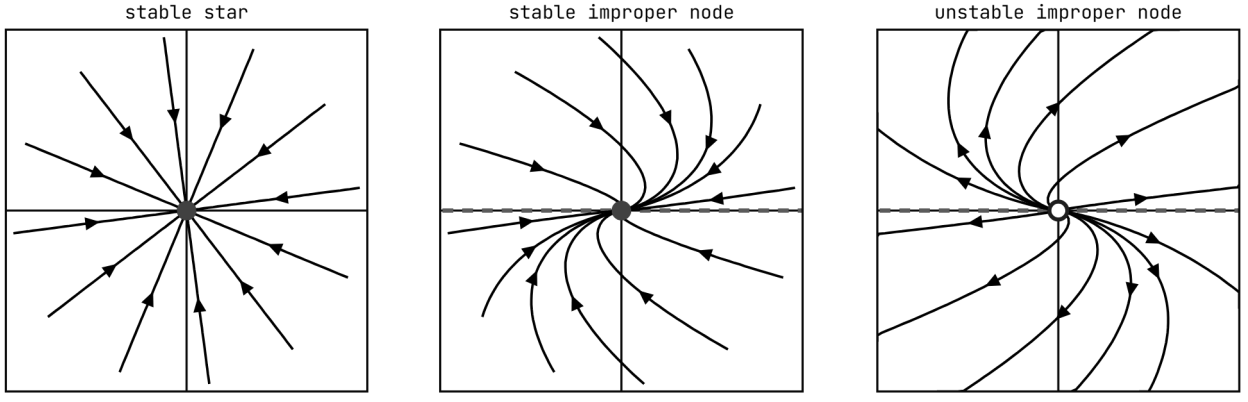


Figure 4.12 (c). *Third row: a repeated eigenvalue (stable star, stable improper node, unstable improper node). Drawn by numerical integration for representative matrices; dashed amber lines are eigenvector directions.*

9.1 The trace–determinant shortcut (complementary)

Since the eigenvalues solve $\lambda^2 - T\lambda + D = 0$ with $T = \operatorname{tr} A = \lambda_1 + \lambda_2$ and $D = \det A = \lambda_1\lambda_2$, the type can be read off T and D without computing the eigenvalues:

- $D < 0$: real eigenvalues of opposite sign – **saddle**.
- $D > 0$ and $T^2 > 4D$: real, same sign as T – **node** (stable if $T < 0$).
- $D > 0$ and $T^2 < 4D$: complex with $\rho = T/2$ – **spiral** (stable if $T < 0$), or **centre** if $T = 0$.
- $T^2 = 4D$: repeated eigenvalue – star or improper node.

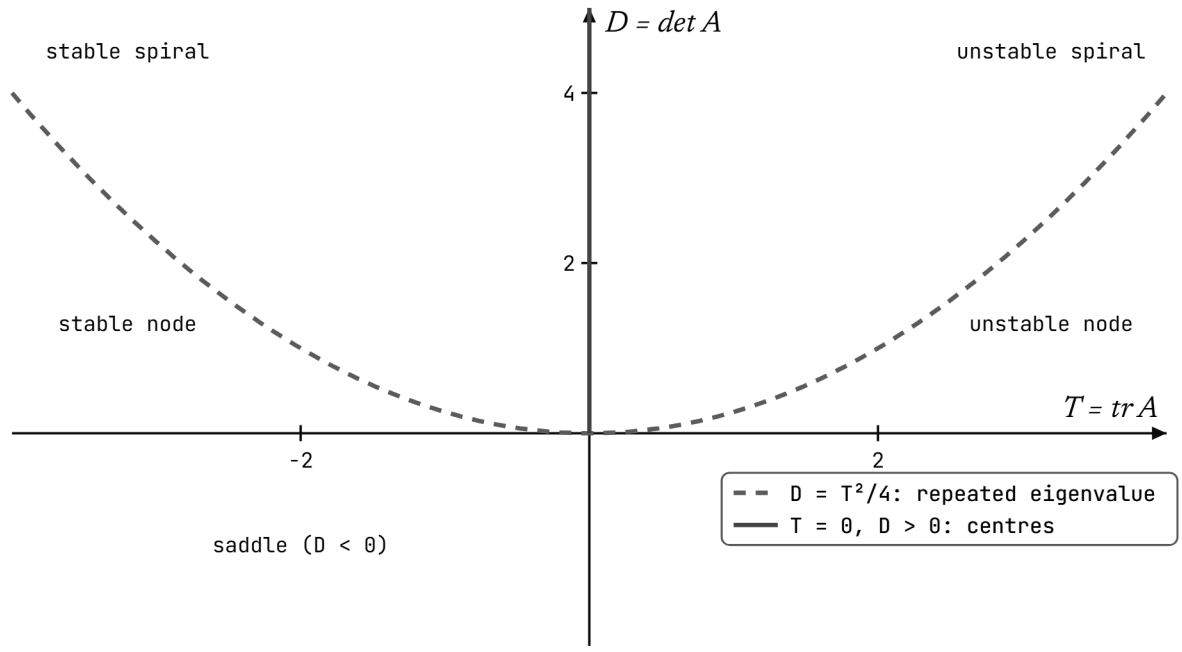


Figure 4.13. *The trace–determinant plane. Since $\lambda^2 - T\lambda + D = 0$, the eigenvalues are real when $T^2 \geq 4D$ (below the dashed parabola, on which they are repeated: stars and improper nodes) and complex above it. This chart is a standard summary that is not in Robinson, who classifies by the eigenvalues directly (Ch. 31).*

THE MASS-SPRING-DAMPER IN THE PHASE PLANE

For $A = \begin{pmatrix} 0 & 1 \\ -\omega^2 & -\lambda \end{pmatrix}$ (Section 1), $T = -\lambda$ and $D = \omega^2 > 0$, and the characteristic equation $\mu^2 + \lambda\mu + \omega^2 = 0$ is the auxiliary equation of Topic 3. The damping regimes are exactly the portrait types:

DAMPING	CONDITION	PHASE PORTRAIT IN $(x, \dot{x})$
none	$\lambda = 0$ ($T = 0$)	centre: energy ellipses $\frac{1}{2}\dot{x}^2 + \frac{1}{2}\omega^2 x^2 = E$
under-damped	$\lambda^2 < 4\omega^2$	stable spiral, clockwise (on $x = 0$, $\dot{x} > 0$ means x increases)
critical	$\lambda^2 = 4\omega^2$	stable improper node ($A \neq \lambda I$ always, since the top-left entry is 0)
over-damped	$\lambda^2 > 4\omega^2$	stable node

The same classification underlies flight dynamics: linearising the longitudinal equations of an aircraft about trimmed flight gives a matrix whose complex eigenvalue pairs are the short-period and phugoid modes, and their real parts decide whether the aircraft returns to trim after a gust.

LECTURE COVERAGE

Sections 1–6 follow page 9 of the class notes and the lectures up to Oct 8 (Ch. 25–28, Sec. 28.4). Sections 7 (complex eigenvalues, Ch. 29, Oct 13) and 8 (a repeated eigenvalue, Ch. 30, Oct 15) follow page 10. Sec. 28.5 (stable and unstable manifolds) and Ch. 31 (summary and Jordan form) are included here so that the topic is complete.